Oscillations

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Simple Harmonic Motion

The study and control of oscillations are two main goals of engineering physics. Pistons in engines of vehicles perform to and fro motion to produce power. This to and fro motion of the pistons is an example of motion that repeats itself, but this motion is different from the periodic motion of the planets around the sun. Here the piston moves to and fro about a mean position. Such a motion is called "oscillatory" or "vibratory" motion.
In this chapter, we will study the simple harmonic motion, periodic motion of spring mass systems and pendulums, physical pendulums and damped and forced oscillating systems.

Periodic Motion.
Motion which is repeating again and again after a regular time interval is known as a periodic motion. The regular time interval after which it repeats is known as the time period of the motion.

Few examples of periodic motion are:
(i) Earth revolves around the Sun once in a time period of 1 year.
(ii) Earth rotates about its own axis in a time period of 24 hours.
(iii) Hour hand, minute hand and second hand of a clock rotate after a time period of 12 hour, 1 hour and 1 minute respectively.

Oscillatory or Vibratory Motion.
A "to and fro" motion or a "back and forth" motion about a fixed point in a fixed time interval is termed as oscillatory or vibratory motion. There are well defined limits on either sides of a mean position (fixed point) in which the body performing the vibration remains confined. There is a slight difference between vibrations and oscillations. For a small frequency, the "back and forth" motion is termed as oscillation while for a high frequency, it is termed as vibration.

Few examples of oscillatory motion are:
(i) Oscillation of a pendulum of clock.
(ii) Oscillation of a mass attached to a vertical spring.
(iii) Vibration of the strings in musical instruments.

Every oscillatory motion is periodic, but every periodic motion need not to be oscillatory. e.g., Motion of planets around sun is periodic but it is not oscillatory because there is no to and fro motion about any fixed point.

Harmonic and Non-harmonic Oscillation.
An oscillation which can be written in terms of a single harmonic function like sine or cosine function is termed as a Harmonic oscillation.

A harmonic oscillation which has a constant amplitude and has a single frequency is known as "Simple Harmonic Motion (SHM)". A SHM is the simplest form of oscillatory motion.

In SHM, the force acting on the oscillating body is directly proportional to its displacement from the mean position.

Also, at any point in its oscillation, this force is directed towards the mean position.

Mathematically, a SHM can be expressed as

$$y = A\sin(\omega t + \phi) \text{ or } y = A\cos(\omega t + \phi).$$

A motion which cannot be expressed in terms of a single harmonic function os termed as non-harmonic oscillation.
It is a combination of two or more than two harmonic oscillations. Mathematically, it can be expressed as-

$$y = A\sin\omega t + B\sin 2\omega t$$

Some Important Definitions

(1) **Time period:** The smallest time interval in which the periodic motion is repeated called the time period.
S.I. unit of time period is second.

(2) **Frequency:** The number of the periodic motions completed in 1 second is known as frequency.
S.I unit of frequency is hertz (Hz).

(3) **Angular Frequency:** The multiplication of the factor 2π with the frequency is known as angular frequency.
Angular frequency $\omega = 2\pi n$
S.I. units of ω is Hz [S.I.].
ω also represents angular velocity. Its unit is rad/sec.

(4) **Displacement:** Displacement is the change in the position wrt time from the mean position in a periodic motion. e.g. In a vertical spring attached with a mass, the mass moves above and below the mean position when it is stretched and released.

(5) **Phase:** The phase of a body in an oscillatory motion expresses the position and the direction of motion of the particle at any instant wrt the mean position.
In oscillatory motion, the phase of a vibrating particle is the argument of sine or cosine function involved to represent the generalized equation of motion of the vibrating particle. $y = a\sin\theta = a\sin(\omega t + \phi_0)$ here, $\theta = \omega t + \phi_0$ = phase of vibrating particle.

(i) **Initial phase:** The phase of the particle at t=0 is termed as the initial phase. In $\theta = \omega t + \phi_0$, when $t = 0$; $\theta = \phi_0$ here, ϕ_0 is the initial phase or angle of epoch.

(ii) **Same phase:** When two particle executing oscillation have a path difference which is an even multiple of (λ / 2) or time interval is an even multiple of (T / 2) or a phase difference which is an even multiple of π, then they are said to be in same phase.

(iii) **Opposite phase:** When the two vibrating particles cross their respective mean positions at the same time moving in opposite directions, then the phase difference between the two vibrating particles is 180°.
Opposite phase means the phase difference between the particle is an odd multiple of π (say π, 3π, 5π, 7π.....) or the path difference is an odd multiple of (λ / 2) (say $\frac{\lambda}{2}, \frac{3\lambda}{2}, ...$) or the time interval is an odd multiple of (T / 2).

(iv) **Phase difference:** If two particles perform S.H.M and their equations are $y_1 = a\sin(\omega t + \phi_1)$ and $y_2 = a\sin(\omega t + \phi_2)$, then phase difference $\Delta\phi = (\omega t + \phi_2) - (\omega t + \phi_1) = \phi_2 - \phi_1$

Displacement in S.H.M.

The displacement of a particle executing S.H.M. at an instant is defined as the distance of particle from the mean position at that instant.

A motion can be considered simple harmonic motion, if the displacement x from the mean position can be expressed as a function of time in the form-

$$x = A\sin(\omega t + \phi) \text{ or } x = A\cos(\omega t + \phi).$$

where, A = Amplitude, ω = Angular frequency, t = Instantaneous time, and ϕ = Initial phase of particle

If the displacement of particle is taken on y-axis then equation of S.H.M. can be given as

$$y = A\cos(\omega t + \phi) \text{ or } y = A\sin(\omega t + \phi)$$

Important points

(i) $y = A\sin\omega t$ when the time is noted from the instant when the vibrating particle is at mean position.

(ii) $y = A\cos\omega t$ when the time is noted from the instant when the vibrating particle is at extreme position.

(iii) $y = A\sin(\omega t \pm \phi)$ when the vibrating particle is leading or lagging by a phase ϕ from the mean position at t=0.

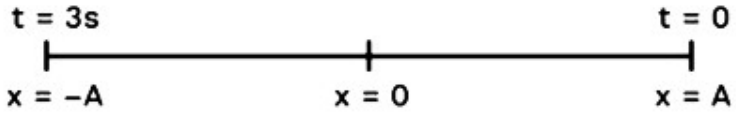
(iv) Direction of displacement is always away from the equilibrium position, particle either is moving away from or is coming towards the equilibrium position.

Q. **For a particle moving as per the equation $x = A\cos\pi t$. The displacement of the particle in 3 s is**

Sol. Given that, $x = A\cos\pi t$

Initially at $t = 0$, $x = A\cos(0) = A$

Finally at $t = 3$ s, $x = A\cos(3\pi) = -A$



We know that displacement is defined as the distance between final and initial positions, so the displacement in 3 s is 2A.

Q. **If the frequency of human heart beat is 1.25 Hz, the number of heart beats in 1 minute is**

Sol. Beat frequency of heart = 1.25 Hz

$\therefore$ Number of beats in 1 minute = $1.25 \times 60 = 75$

Q. **A simple harmonic motion is represented by, $x(t) = \sin^2\omega t - 2\cos^2\omega t$. The angular frequency of oscillation is given by**

Sol. $x(t) = \sin^2\omega t - 2\cos^2\omega t$

$$= \left(\frac{1-\cos 2\omega t}{2}\right) - (1+\cos 2\omega t) = -\frac{1}{2} - \frac{3}{2}\cos 2\omega t$$

So, the frequency of oscillation is 2ω.

Q. **The ratio of amplitudes of two simple harmonic motions represented by the equations $y_1 = 5\sin\left(2\pi t + \frac{\pi}{4}\right)$ and $y_2 = 2\sqrt{2}\left(\sin 2\pi t + \cos 2\pi t\right)$ is**

Sol. Given, $y_1 = 5\sin\left(2\pi t + \frac{\pi}{4}\right)$

Comparing with standard equation, we get $A_1 = 5$

and $$y_2 = 2\sqrt{2}\left(\sin 2\pi t + \cos 2\pi t\right) = 2\sqrt{2} \times \sqrt{2}\left(\frac{1}{\sqrt{2}} \times \sin 2\pi t + \frac{1}{\sqrt{2}} \times \cos 2\pi t\right)$$

$$y_2 = 4\left(\sin 2\pi t \,.\cos\frac{\pi}{4} + \cos 2\pi t.\,\sin\frac{\pi}{4}\right) = 4\sin\left(2\pi t + \frac{\pi}{4}\right)$$

Then, $A_2 = 4$

$\therefore$ $$\frac{A_1}{A_2} = \frac{5}{4}$$

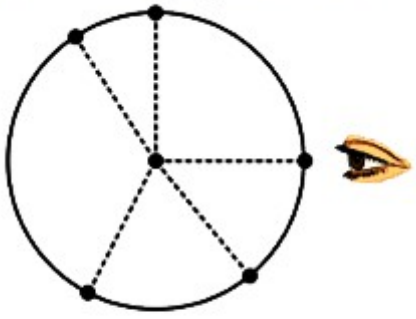
SIMPLE HARMONIC MOTION AND UNIFORM CIRCULAR MOTION

The projection or the shadow of the uniform circular motion on the diameter of the circle is a Simple Harmonic Motion.

Consider a ball is attached to the end of a string the other end of which is fixed and it is placed in the horizontal plane. Lets consider the ball is undergoing uniform circular motion in the plane.

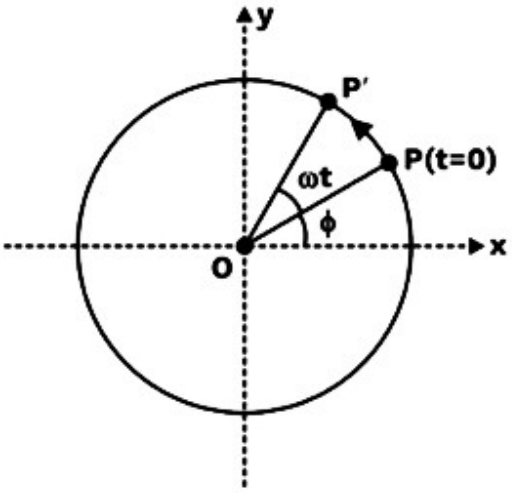
If the motion of the ball is observed from one side, then it appears that the ball is performing "to and fro" motion along the diameter of the circle with the centre as the mean position.

Other way to observe this to and fro motion is to see the shadow of the ball on a vertical wall which is perpendicular to the horizontal plane in which the ball is rotating.



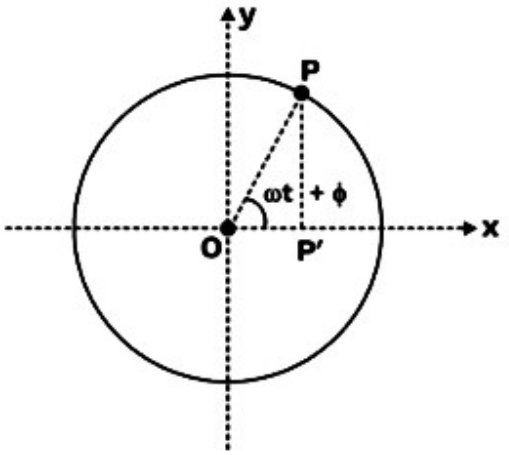
Consider a particle P to be moving in uniform circular motion in a circle of radius A with angular speed ω in anti-clockwise direction.

$\overrightarrow{(OP)}$ is the initial position vector of the particle making an angle ϕ with positive direction of x-axis at t = 0.



In time t, it covers a further angle ωt and its position vector makes an angle $(\omega t + \phi)$ with positive x-axis.

Consider the projection of position vector OP on x-axis. Let this be OP′.



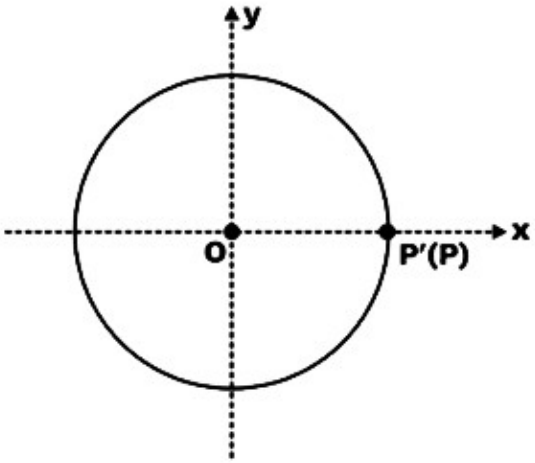
From the figure, we can write that,

$$x(t) = A \cos(\omega t + \phi)$$

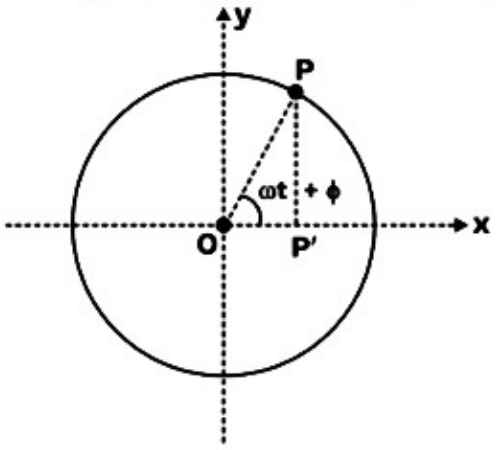
which is the defining equation of S.H.M.

(a) Consider that at t = 0, $\omega t + \phi = 0$. Then the particle will be at P. Its projection on x-axis is at the same point.

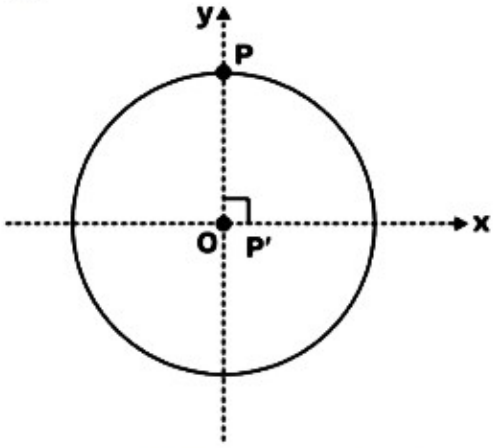
$$OP = OP' = A \cos 0° = A.$$



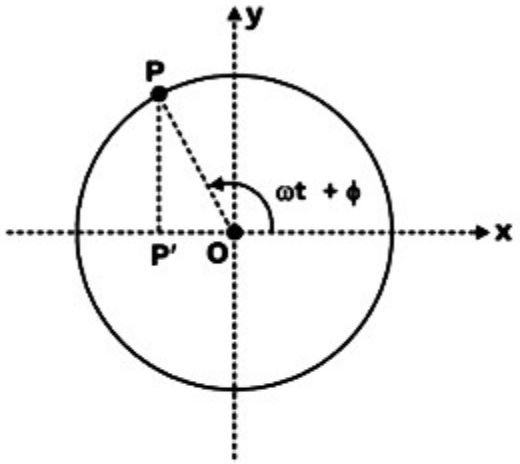
(b) When the position vector $\overline{(OP)}$ makes an angle $(\omega t + \phi)$ with positive x-axis at time t, then its projection on x-axis is given as, OP′ = A cos $(\omega t + \phi)$.



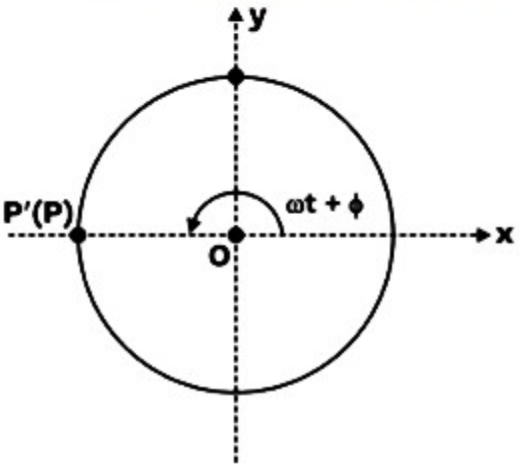
(c) For $(\omega t + \phi)$ as 90°, the projection of P on x-axis is at the centre O i.e., OP = O as OP′ = A cos $(\omega t + \phi)$ = A cos 90° = 0.



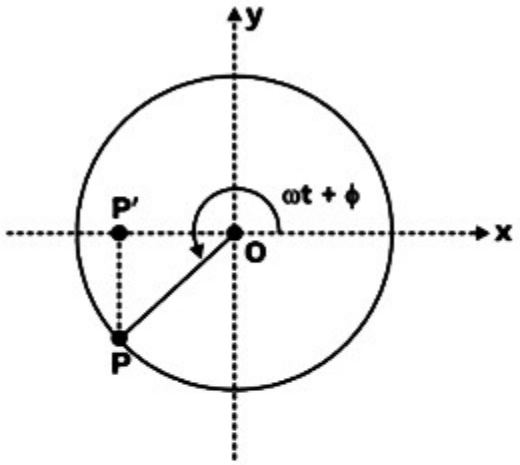
(d) For $(\omega t + \phi) > 90°$ i.e. the particle in 2nd quadrant, the projection of the particle on the x-axis is OP′ = A cos $(\omega t + \phi)$. As the phase is more than 90°, so the projection has a negative value.



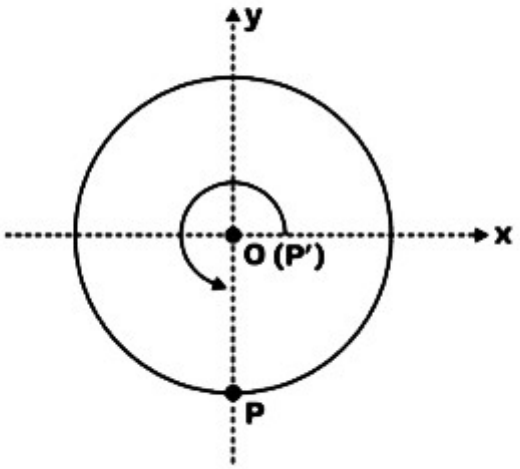
(e) For $(\omega t + \phi) = 180°$, the projection on x-axis is $OP' = A \cos 180° = -A$.



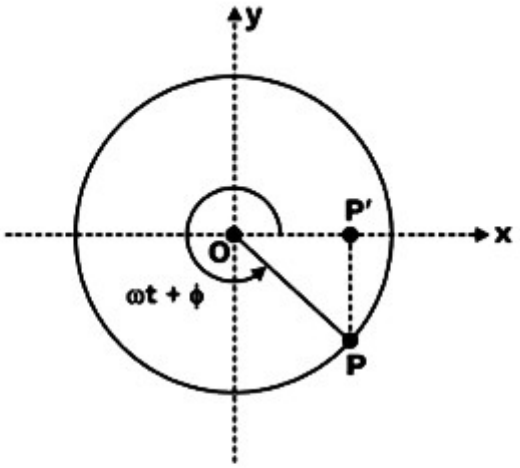
(f) For $(\omega t + \phi) > 180°$, the particle is in 3rd quadrant and Its projection on x-axis is again negative.



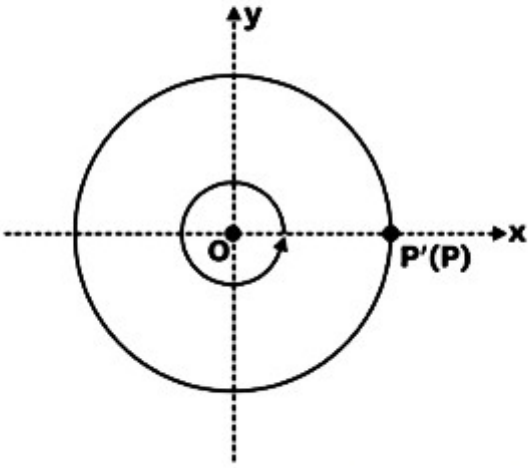
(g) For $(\omega t + \phi) = 270°$, the projection of the particle is on the x-axis is at centre O.



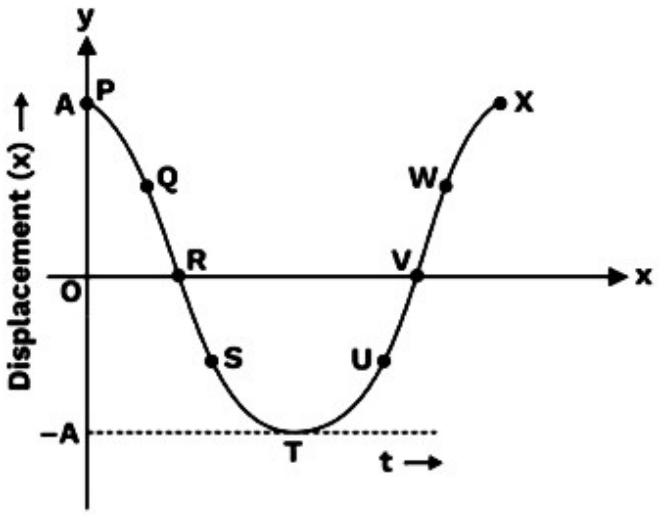
(h) For $(\omega t + \phi) > 270°$, the particle is in the 4th quadrant. The projection on the x-axis is positive.



(i) For $(\omega t + \phi) = 360°$, particle comes back to the initial position. Its projection on x-axis is same as that in the start.



Thus, we can see that when a particle moves in uniform circular motion, then its projection on the diameter of the circle follows simple harmonic motion



P in above figure corresponds to projection OP' at $(\omega t + \phi) = 0$

Q in above figure corresponds to projection OP' with $0 < (\omega t + \phi) < 90°$

R in above figure corresponds to projection OP' at $(\omega t + \phi) = 90°$

S in above figure corresponds to projection OP' with $90° < (\omega t + \phi) < 180°$

T in above figure corresponds to projection OP' with $(\omega t + \phi) = 180°$

U in above figure corresponds to projection OP' with $180° < (\omega t + \phi) < 270°$

V in above figure corresponds to projection OP' with $(\omega t + \phi) = 270°$

W in above figure corresponds to projection OP' with $270° < (\omega t + \phi) < 360°$

X in above figure corresponds to projection OP' with $(\omega t + \phi) = 360°$

The circle is called is reference circle and the particle is called reference particle.

The projection of the particle in uniform circular motion can be taken on either of the diameters.

For the projection on the y-axis,

$y = A \sin(\omega t + \phi)$, which is again an SHM equation having a phase difference of $\pi/2$ with the projection taken on the x-axis.

Note: Although there is a connection between the motion of a particle in circular path and its projection as SHM, but the forces acting on a particle in SHM and in circular motion are different. In SHM, a restoring force always acting towards the mean position is present while in circular motion, there is a centripetal force always acting towards the centre of the circle.

Q.

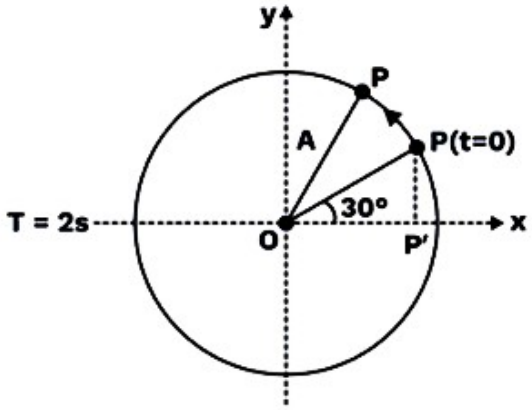


Fig. (a)

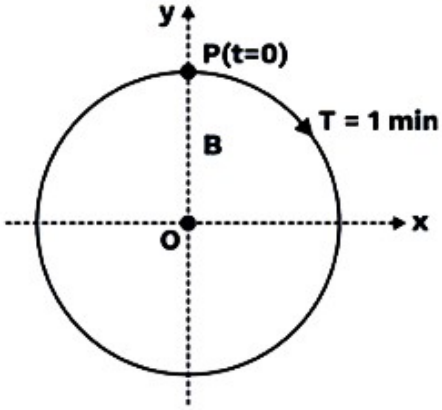


Fig. (b)

In the above figures, the revolution period, initial position, revolution sense and radius of the circle are shown for two circular motions. Obtain the SHMs of the x-projection of the radius vector of the rotating particle P in each case.

Sol. (a) The position vector OP is making an angle of $30° = \frac{\pi}{6}$ rad with the positive x-axis at t = 0.

We know that, $\omega = \frac{2\pi}{T}$.

So, we can write the projection of OP on x-axis as, $x(t) = A\cos\left(\frac{2\pi}{T}t + \frac{\pi}{6}\right)$

For t = 2s,

$$x(t) = A\cos\left(\frac{2\pi}{2}t + \frac{\pi}{6}\right)$$

This is an SHM of amplitude A, period 2s, and an initial phase $= \frac{\pi}{6}$.

(b) The position vector OP is making an angle of $90° = \frac{\pi}{2}$ with the x-axis at t=0.

So, we can write the projection of OP on x-axis as,

$$x(t) = B\sin\left(\frac{2\pi}{T}t\right)$$

For T = 1 min = 60 s,

$$x(t) = B\sin\left(\frac{\pi}{30}t\right)$$

Q. **Calculate the time the particle takes in moving from x = 0 to $x = \frac{A}{2}$ where A is the amplitude of the SHM**

Sol. Let the SHM equation to be, $x = A \sin \omega t$

Then, for $x = \frac{A}{2}$, $\frac{A}{2} = A \sin \omega t$

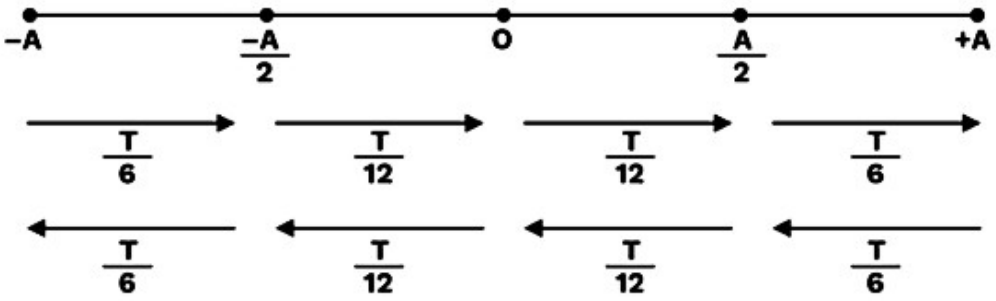
$$\sin \frac{\pi}{6} = \sin \frac{2\pi}{T} t$$

$$\frac{\pi}{6} = \frac{2\pi}{T} t$$

$$\Rightarrow t = \frac{T}{12}$$

Note : Time taken by the particle in going,

(1) From x = –A to $x = -\frac{A}{2}$ is same as that in $x = -\frac{A}{2}$ to x = –A, and this is $\frac{T}{6}$.



(2) From $x = -\frac{A}{2}$ to x = 0 is same as that in x = 0 to $x = -\frac{A}{2}$ and this is $\frac{T}{12}$.

(3) From x = 0 to $x = \frac{A}{2}$ is same as that in $x = \frac{A}{2}$ to x = 0 and this is $\frac{T}{12}$.

(4) From $x = \frac{A}{2}$ to x = A is same as that in x = A to $x = \frac{A}{2}$ and this is $\frac{T}{6}$.

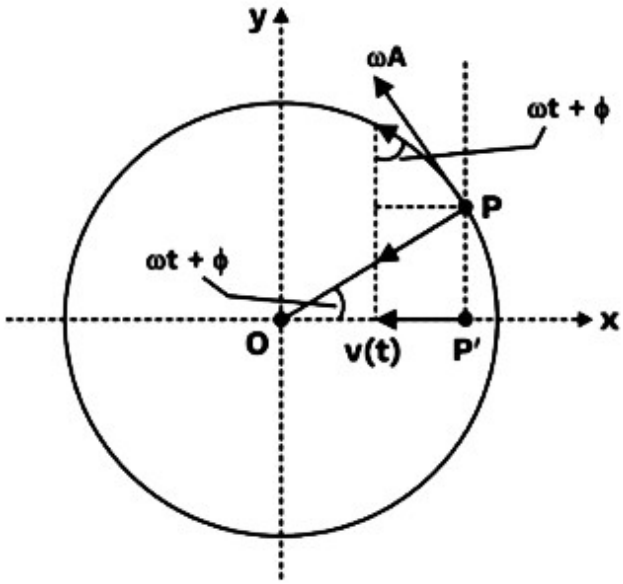
These results can be generalized as follows:

Time (from $x = x_1$ to $x = x_2$) = time (from $x = x_2$ to $x = x_1$)
= time (from $x = -x_1$ to $x = -x_2$)
= time (from $x = -x_2$ to $x = -x_1$)

VELOCITY AND ACCELERATION IN SIMPLE HARMONIC MOTION

In uniform circular motion, the speed v is product of the angular speed (ω) and radius of the circular path (A).

$v = \omega A$



The projection of the velocity of reference particle gives the velocity of the point P'.

The velocity $\vec{v}$ at any time t is along the tangent drawn on the circle.

From the geometry of the given figure, it can be seen that the velocity of the projection particle P' at time t is

$v(t) = -\omega A \sin(\omega t + \phi)$

where the –ve sign shows that direction of v(t) is opposite to the +ve direction of x-axis.

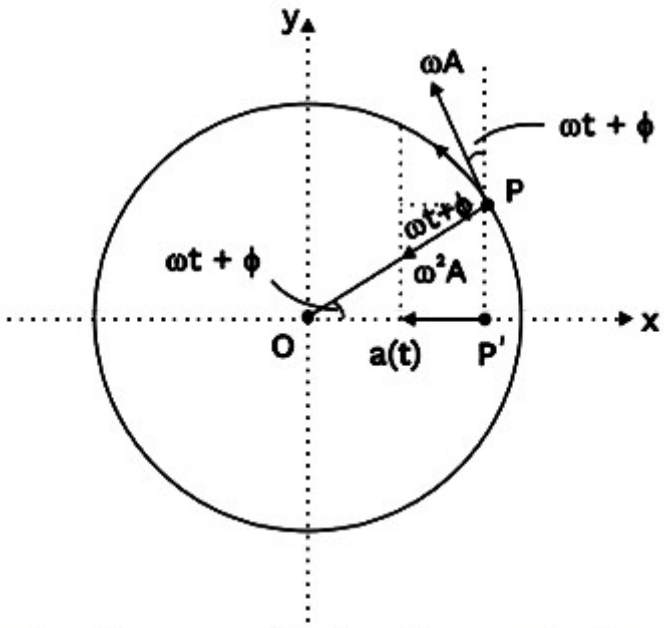
Above equation gives the instantaneous velocity of a particle executing S.H.M. where displacement is given by equation $x(t) = A \cos(\omega t + \phi)$.

This equation can also be obtained by differentiating the displacement equation w.r.t. time,

$$v(t) = \frac{d}{dt} x(t)$$

Reference circle method can be used to find the instantaneous acceleration as well.

We know that, centripetal acceleration, $a_c = \frac{v^2}{A} = \omega^2 A$ (towards PO).



The projection of this acceleration on x-axis gives the acceleration of P'.

$a(t) = -\omega^2 A \cos(\omega t + \phi)$

$a(t) = -\omega^2 x(t)$

This equation provides the acceleration of a particle in SHM.

The above equation can also be obtained by differentiating velocity w.r.t. time:

$$a(t) = \frac{d}{dt} v(t)$$

Note :

(1) From acceleration equation, we can see that acceleration in SHM is proportional to displacement.

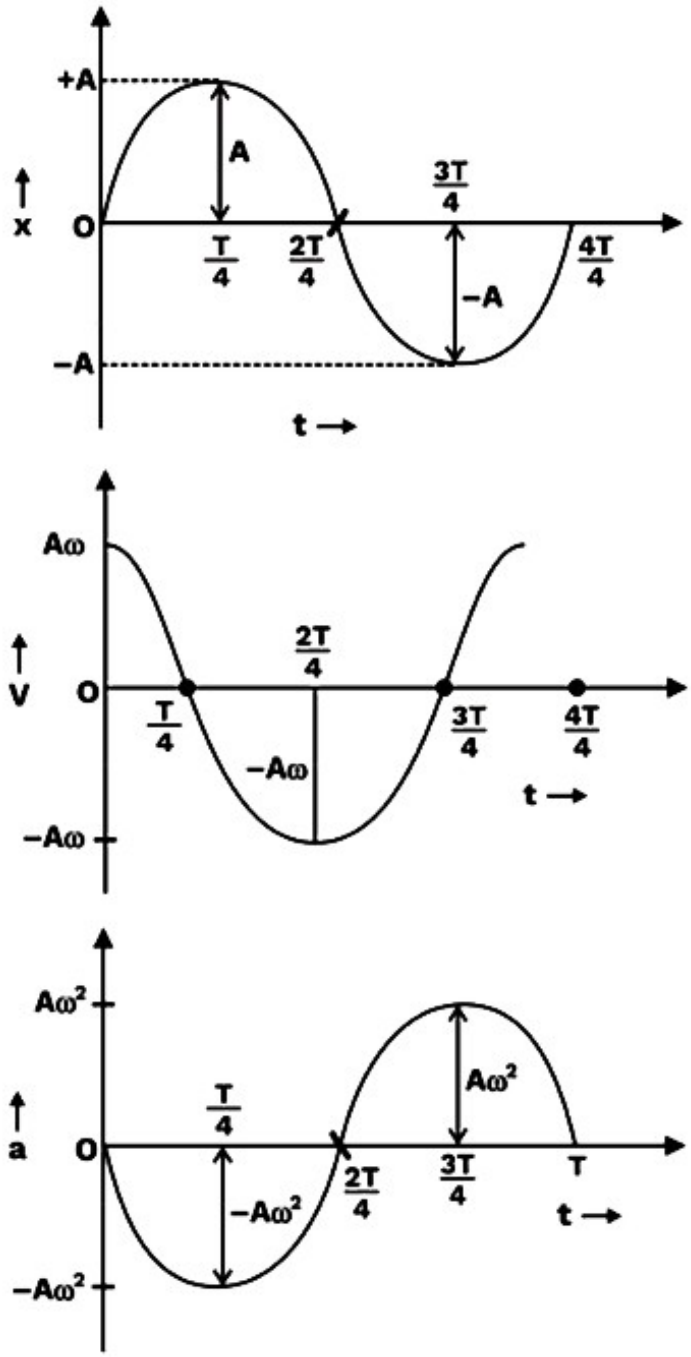
(2) The acceleration a(t) is always directed towards the mean position.

$x(t) = A \sin \omega t$

$$v(t) = \frac{dx}{dt} = \omega A \cos \omega t = A\omega \sin\left(\omega t + \frac{\pi}{2}\right)$$

$$a(t) = \frac{dv}{dt} = -\omega^2 A \sin \omega t = A\omega^2 \sin(\omega t + \pi)$$

The equation for $x(t) = A \sin \omega t$ and corresponding v(t) and a(t) can be graphically drawn as:



Below table gives the values of x, v and t at different time instants.

Time, t	0	T/4	2T/4	3T/4	T
Displacement, x	0(min)	A(max)	0(min)	−A(max)	0(min)
Velocity, v	Aω(max)	0(min)	−Aω(max)	0(min)	Aω(max)
Acceleration, a	0(min)	$-A\omega^2$(max)	0(min)	$A\omega^2$(max)	0(min)

Note :

(a) In SHM, the time period of displacement, velocity and acceleration is same..

(b) The velocity amplitude is ω times the displacement amplitude.

(c) The acceleration amplitude is ω^2 times the displacement amplitude.

(d) In S.H.M., velocity leads displacement by phase angle $\pi/2$.

(e) In S.H.M., acceleration leads velocity by phase angle $\pi/2$.

(f) In S.H.M., acceleration leads displacement by phase angle of π.

For a simple SHM represented by equation,

$x = A \sin \omega t$

Velocity in S.H.M. at instant t is,

$$v = \frac{dx}{dt} = A\omega \cos \omega t$$

$$\boxed{v = A\omega \cos \omega t}$$

$$= A\omega\sqrt{1 - \sin^2 \omega t}$$

$$= A\omega\sqrt{1 - \frac{x^2}{A^2}}$$

or $\boxed{v = \omega\sqrt{A^2 - x^2}}$

The above equation gives the relation between velocity and displacement in SHM.

When the displacement is zero ($x = 0$), i.e., when it passes through its equilibrium position, velocity is maximum ($v_{max} = A\omega$) and when the displacement is maximum ($x = A$), velocity is zero ($v = 0$).

Also, the acceleration will be,

$$a = \frac{dv}{dt} = -A\omega^2 \sin \omega t$$

$$a = -\omega^2 x$$

$$a \propto -x$$

Thus, acceleration of SHM is proportional to the displacement x and its direction is opposite to the direction of displacement from the equilibrium.

Q. **A particle executes SHM such that its displacement from x = 0 is given by x = A sin ωt. If the speeds of the particle at $x = x_1$ and $x = x_2$ are v_1 are v_2 respectively, then find the time period of oscillations.**

Sol. For the displacement equation given by x = A sin ωt, the velocity and position are related as $v = \omega\sqrt{A^2 - x^2}$.

At $x = x_1$, $v = v_1$, and

At $x = x_2$, $v = v_2$. So, we have

$$v_1 = \omega\sqrt{A^2 - x_1^2}$$

$$v_2 = \omega\sqrt{A^2 - x_2^2}$$

Squaring and subtracting, we get

$$v_1^2 - v_2^2 = \omega^2[x_2^2 - x_1^2]$$

$$\Rightarrow \omega = \sqrt{\frac{v_1^2 - v_2^2}{x_2^2 - x_1^2}}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{v_1^2 - v_2^2}}$$

ACCELERATION

If v = velocity of particle then $\frac{dv}{dt}$ = acceleration given by

$$a = -A\omega^2 \sin(\omega t + \phi_0)$$

or $a = -\omega^2 x$ $[\because x = A \sin(\omega t + \phi_0)]$

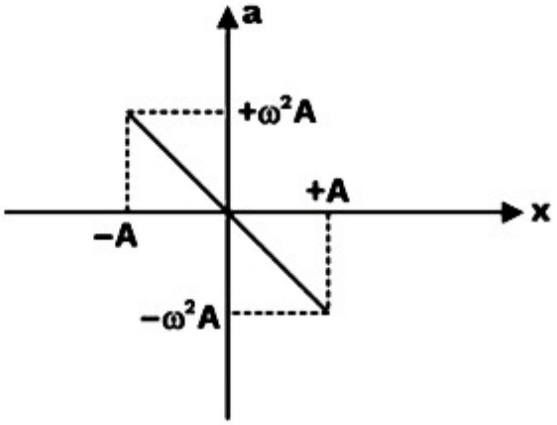
Also, $a = -a_0 \sin(\omega t + \phi_0)$

Where $a_0 = \omega^2 A$ is known as maximum acceleration.

It can also be written as,

$$a = a_0 \sin[(\omega t + \phi_0) + \pi]$$

Variation of acceleration (a) with displacement (x) can be plotted as:



Q. **Find whether the function (sinωt + cosωt) is Simple harmonic motion and find its period.**

Sol. $\sin\omega t + \cos\omega t$

$$= \sin\omega t + \sin\left(\frac{\pi}{2} - \omega t\right)$$

$$= 2\sin\left(\frac{\pi}{4}\right)\cos\left(\omega t - \frac{\pi}{4}\right) \qquad \left(\text{using } \sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}\right)$$

$$= 2 \times \frac{1}{\sqrt{2}}\cos\left(\omega t - \frac{\pi}{4}\right)$$

$$= \sqrt{2}\cos\left(\omega t - \frac{\pi}{4}\right)$$

This function represents a SHM having period $T = \frac{2\pi}{\omega}$ and phase angle $\left(-\frac{\pi}{4}\right)$ or $\left(\frac{7\pi}{4}\right)$.

Q. **Find whether the given functions represent "simple harmonic motion", "oscillatory but not SHM" or "periodic but not oscillatory" motions. Also find the time period, if the motion is periodic.**

(A) $x = a\sin\omega t + b\cos\omega t$

(B) $x = \sin\omega t + \sin 2\omega t + \sin 3\omega t$

(C) $x = Ae^{-\lambda t}\sin\omega t$

Sol. (a) $x = a \sin \omega t + b \cos \omega t$

$$\frac{dx}{dt} = a\omega \cos \omega t - b\omega \sin \omega t$$

$$\frac{d^2x}{dt^2} = -a\omega^2 \sin \omega t - b\omega^2 \cos \omega t$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

So, $x = a \sin \omega t + b \cos \omega t$ represents SHM.

Now as x is a function of time

$x = a \sin \omega t + b \cos \omega t$.

As $\sin(\theta + 2n\pi) = \sin \theta$ and $\cos(\theta + 2m\pi) = \cos \theta$

So, we can write

$x = a \sin(\omega t + 2n\pi) + b \cos(\omega t + 2n\pi)$

$$x = a \sin \omega\left(t + \frac{2n\pi}{\omega}\right) + b \cos \omega\left(t + \frac{2n\pi}{\omega}\right)$$

$\therefore$ The value of x at t and $t + \frac{2n\pi}{\omega}$ is same.

So, x is a periodic function with time period $T = \frac{2\pi}{\omega}$.

(b) $x = \sin \omega t + \sin 2\omega t + \sin 3\omega t$

$$\frac{dx}{dt} = \omega \cos \omega t + 2\omega \cos 2\omega t + 3\omega \cos 3\omega t$$

$$\frac{d^2x}{dt^2} = -\omega^2 \sin \omega t - 4\omega^2 \sin \omega t - 9\omega^2 \sin 3\omega t$$

$$\frac{d^2x}{dt^2} = -\omega^2(\sin \omega t + 4 \sin 2\omega t + 9 \sin 3\omega t)$$

So, $\frac{d^2x}{dt^2} \neq -\omega^2 x$

So, the given function does not represent SHM.

Let $T = \frac{2\pi}{\omega}$

$x(t + nT) = \sin \omega(t + nT) + \sin 2\omega(t + nT) + \sin 3\omega(t + nT)$

$= \sin(\omega t + 2n\pi) + \sin(2\omega t + 4n\pi) + \sin(3\omega t + 6n\pi)$ [As $\omega T = 2\pi$]

$= \sin \omega t + \sin 2\omega t + \sin 3\omega t$

$\Rightarrow \quad x(t + nT) = x(t)$

So, the given function is periodic with time period $T = \frac{2\pi}{\omega}$.

(c) $x = Ae^{-\lambda t} \sin \omega t$

$$\frac{dx}{dt} = A\omega e^{-\lambda t} \cos \omega t - A\lambda e^{-\lambda t} \sin \omega t$$

$$\frac{d^2x}{dt^2} = -A\omega^2 e^{-\lambda t} \sin \omega t - 2A\omega\lambda e^{-\lambda t} \cos \omega t + A\lambda^2 e^{-\lambda t} \sin \omega t$$

As, $\frac{d^2x}{dt^2} \neq -\omega^2 x$

Hence the given function does not represent SHM.

Also it can be shown that

$x(t + nT) \neq x(t)$, so the given function is not periodic.

Q. **Find the amplitude for SHM given by equation $x = A\sin\omega t + B\cos\omega t$.**

Sol. Given equation is $x = A \sin\omega t + B \cos\omega t$

$$x = \left(\frac{A}{\sqrt{A^2 + B^2}} \sin \omega t + \frac{B}{\sqrt{A^2 + B^2}} \cos \omega t\right)\sqrt{A^2 + B^2} \quad \ldots(i)$$

Let $\frac{A}{\sqrt{A^2 + B^2}} = \cos\alpha, \frac{B}{\sqrt{A^2 + B^2}} = \sin\alpha$

then from equation (i), we get

$$x = (\cos\alpha \sin\omega t + \sin\alpha \cos\omega t)\sqrt{A^2 + B^2}$$

$$x = \sqrt{A^2 + B^2} \sin(\omega t + \alpha)$$

Here, amplitude is $\sqrt{A^2 + B^2}$

And, $\tan\alpha = \frac{B}{A} \Rightarrow \alpha = \tan^{-1}\left(\frac{B}{A}\right)$

Q. **Find the phase difference between two SHMs given by** $x = a\sin\left(\omega t + \frac{\pi}{6}\right)$ **and** $x' = a\cos\omega t$.

Sol. Phase from the equation

$$x = a\sin\left(\omega t + \frac{\pi}{6}\right) \text{ is } \phi_1 = \omega t + \frac{\pi}{6}$$

Phase from the equation

$$x' = a\cos\omega t = a\sin\left(\omega t + \frac{\pi}{2}\right) \text{ is } \phi_2 = \omega t + \frac{\pi}{2}$$

Phase difference,

$$\Delta\phi = \phi_2 - \phi_1$$

$$= \left(\omega t + \frac{\pi}{2}\right) - \left(\omega t + \frac{\pi}{6}\right)$$

$$= \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6}$$

$$= \frac{2\pi}{6} = \frac{\pi}{3}$$

Q. **Find the initial phase for the SHM given by** $y = 0.5\cos\left(100t + \frac{3\pi}{4}\right)$ **m.**

Sol. The phase of the given equation is

$$\phi = 100t + \frac{3\pi}{4},$$

So initial phase (t = 0) is,

$$\phi_0 = (100 \times 0) + \frac{3\pi}{4} = \frac{3\pi}{4}.$$

Q. **Find the amplitude of oscillation for a particle executing SHM along a straight line. Given that the particle starts from rest, travels a distance A in first second and a distance B in next second in same direction.**

Sol. The particle starts from rest, so it starts from the extreme position.
So, we can write, $x = a\cos(\omega t)$, where "a" is amplitude of oscillation.
Given that at $t = 1$ s, $x = a - A$

$\Rightarrow a - A = a\cos\omega$ ...(i)

And, at $t = 2$s, $x = a - (A + B)$

$\Rightarrow a - (A + B) = a\cos 2\omega$ ...(ii)

$$\text{Now, } \cos\omega = \frac{a - A}{a}; \quad \cos 2\omega = 2\cos^2\omega - 1 = 2\left(\frac{a - A}{a}\right)^2 - 1$$

$$\cos 2\omega = \frac{2\left(a^2 + A^2 - 2aA\right) - a^2}{a^2} = \frac{a^2 + 2A^2 - 4aA}{a^2}$$

$$\frac{a - A - B}{a} = \frac{\left(a^2 + 2A^2 - 4aA\right)}{a^2}$$

Then, $3aA - aB = 2A^2$

$$a = \left(\frac{2A^2}{3A - B}\right)$$

Q. **Find the phase difference when two particles cross each other while moving in opposite directions when their displacement is $\frac{1}{\sqrt{2}}$ times the amplitude. The particles are performing SHM with same frequency and amplitude in a straight line.**

Sol. Consider the equation of motion of the particles is

$y_1 = A \sin \omega t$,and $y_2 = A \sin(\omega t + \phi)$

For $y_1 = \frac{A}{\sqrt{2}}$ and $y_2 = \frac{A}{\sqrt{2}}$ we have

$\frac{A}{\sqrt{2}} = A \sin \omega t$ and $\frac{A}{\sqrt{2}} = A \sin(\omega t + \phi)$

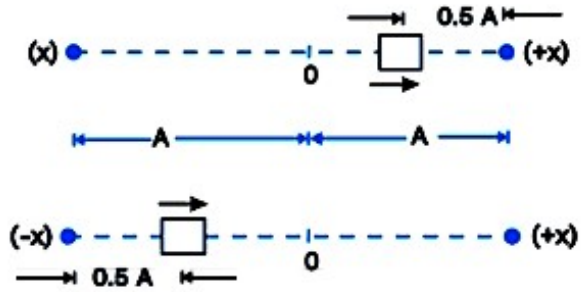
$\omega t = \frac{\pi}{4}$ and $\omega t + \phi = \frac{\pi}{4}$ or $\frac{3\pi}{4}$

As the two particles are not in same phase,

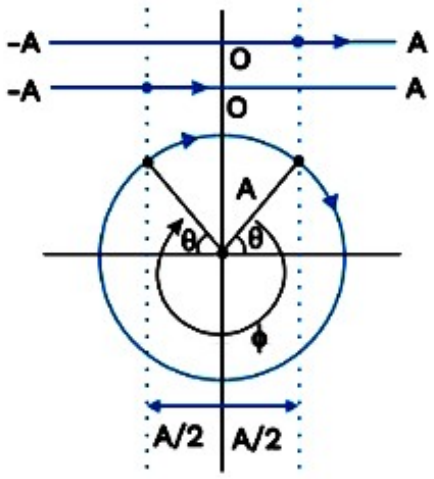
$$\omega t + \phi = \frac{3\pi}{4}$$

$$\Rightarrow \phi = \frac{3\pi}{4} - \frac{\pi}{4} = \frac{\pi}{2}$$

Q. **Find the phase difference between the particles shown in the figure below.**



Sol.



$$\cos\theta = \frac{A/2}{A} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Total phase difference between them, $\phi = \pi + 2\left(\frac{\pi}{3}\right) = \frac{5\pi}{3}$.

Q. **The maximum separation between two particles performing SHM in same straight line with same frequency and same amplitude is 3a/2, where a is the amplitude. What is the displacement when they have same displacement from mean position?**

Sol. Consider that, $x_1 = a\sin\omega t$, $x_2 = a\sin(\omega t + \phi)$

Maximum separation $= |x_2 - x_1|_{max} = 2a\sin\frac{\phi}{2} = \frac{3a}{2} \Rightarrow \sin\frac{\phi}{2} = \frac{3}{4}$

For same displacement of the particles,

$x_1 = x_2$

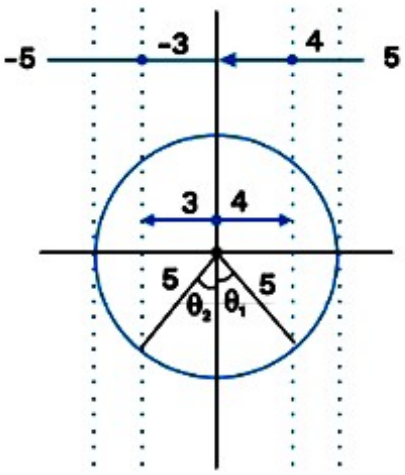
$\Rightarrow a\sin\omega t = a\sin(\omega t + \phi)$

$\Rightarrow \pi - \omega t = \omega t + \phi \Rightarrow \omega t = \frac{\pi - \phi}{2}$

$\Rightarrow x_1 = a\sin\left(\frac{\pi - \phi}{2}\right) = a\cos\frac{\phi}{2} = \frac{a\sqrt{7}}{4}$

Q. **Find the minimum time taken by a particle to move from x= 4 cm to x= -3 cm if the particle is performing SHM in a straight line with x=0 as mean position. The amplitude is 5 cm and time period is 20 s**

Sol.



$$\omega = \frac{2\pi}{T} = \frac{2\pi}{20} = \frac{\pi}{10}$$

From figure $\theta_1 + \theta_2 = 53^0 + 37^0 = 90^0$ or $\frac{\pi}{2}$

$$\theta = \omega t \Rightarrow \frac{\pi}{2} = \frac{\pi}{10} t \Rightarrow t = 5\,\text{sec}$$

FORCE LAW FOR SIMPLE HARMONIC MOTION

By Newton's second law, F = ma

Also, in SHM, $a = -\omega^2 x$

So, force, $F = -m\omega^2 x = -kx$, where $k = m\omega^2$.

Above relation can be written as a second order differential equation,

$m\frac{d^2x}{dt^2} + kx = 0$, wwhich is a standard equation form showing motion is simple harmonic.

This gives, $\omega = \sqrt{\frac{k}{m}}$

Similar to acceleration, the force is also always directed towards the mean position. That's why it is also termed restoring force.

Now, we can write, $\omega = \frac{2\pi}{T} = \sqrt{\frac{k}{m}}$

So, time period, $T = 2\pi\sqrt{\frac{m}{k}}$

In above equation, m represents the inertia factor of the oscillating system and k represents the spring factor of the system.

So, $T = 2\pi\sqrt{\frac{\text{Inertia factor}}{\text{Spring factor}}}$ and, frequency, $\nu = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{\text{Spring factor}}{\text{Inertia factor}}}$

Linear SHM

In linear SHM, spring factor is the restoring force per unit displacement and the inertia factor is the mass of the oscillating body.

So, in linear SHM, $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{m}{\left(\frac{F}{s}\right)}} = 2\pi\sqrt{\frac{m \times s}{m \times a}} = 2\pi\sqrt{\frac{s}{a}}$, where s is the displacement.

Angular SHM

Similar to the linear SHM, in angular SHM, the displacement is angular displacement and restoring torque is directly proportional to angular displacement from mean position.

$$\vec{\tau} \propto -\left(\vec{\theta}\right)$$

$\vec{\tau} = -C\left(\vec{\theta}\right)$, where C is known as torsional constant.

We can write, $\tau = I\alpha = I\frac{d^2\theta}{dt^2}$, where I is the moment of inertia.

So, $I\frac{d^2\vec{\theta}}{dt^2} = -C\vec{\theta}$

$$\frac{d^2\vec{\theta}}{dt^2} + \frac{C}{I}\vec{\theta} = 0$$

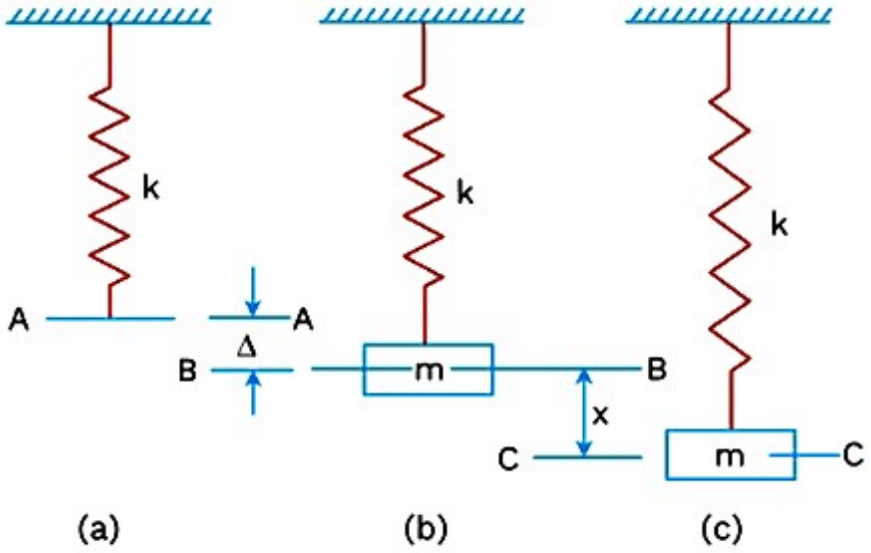
This is the differential equation for angular SHM with $\omega^2 = \frac{C}{I} \Rightarrow \omega = \sqrt{\frac{C}{I}} \Rightarrow T = 2\pi\sqrt{\frac{I}{C}}$

The natural frequency of an oscillating system can be found by any of the following methods.

1. Equilibrium method

At the equilibrium, net force and net torque on the oscillating system is zero.
Figure (a) shows a spring suspended vertically from a rigid support with its free end at A-A.

If a mass m is suspended from the free end, the spring is stretched by a distance Δ and B-B becomes the equilibrium position Figure (b). Thus, Δ is the static deflection of the spring under the weight of the mass m.



Let k = stiffness of the spring under the weight of the mass m.

In the static equilibrium position, upward force = downward force

$k \times \Delta = mg$

Now, on pulling the mass m down by small distance x, the forces on the mass are

Inertia force = ma (upwards)

spring force (restoring force) = kx (upwards)

(x is downward and thus velocity (v) and acceleration (a) are also downwards)

As the sum of the inertia force and the external force on the body in any direction is to be zero, so,

$ma + kx = 0$

If the mass is released, it will start oscillating above and below the equilibrium position. The oscillation will continue forever if there is no frictional resistance to the motion.

The above equation can be written as $a + \left(\frac{k}{m}\right)x = 0$

This equation is similar to $a + \omega_n^2 x = 0$, the solution of which is given by

$$x = A\sin\omega t + B\cos\omega t$$

2. Energy Method

In a conservative system (a system with no damping), the total mechanical energy, i.e., the sum of the kinetic and the potential energies, remains constant and therefore,

$$\frac{d}{dt}(KE + PE) = 0$$

We have $KE = \frac{1}{2}mv^2$

and PE = mean force × displacement $= \frac{0 + kx}{2} \times x = \frac{kx^2}{2}$

$\therefore$ $$\frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{kx^2}{2}\right) = 0$$

or $$\frac{1}{2}m \times 2v\left(\frac{dv}{dt}\right) + \frac{1}{2}k \times 2x\left(\frac{dx}{dt}\right) = 0 \quad \text{or} \quad ma + kx = 0 \quad \text{or} \quad \omega = \sqrt{\frac{k}{m}}$$

Displacement, velocity and acceleration

This displacement of the mass m from the mean position at any instant is

$$x = A\sin(\omega t + \phi)$$

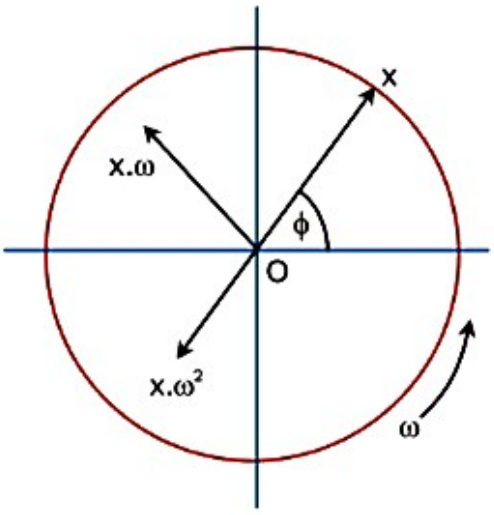
Also velocity,

$$v = \frac{dx}{dt} = A\omega\cos(\omega t + \phi) = A\omega\sin\left[\frac{\pi}{2} + (\omega t + \phi)\right]$$

and acceleration,

$$a = \frac{dv}{dt} = -A\omega^2\sin(\omega t + \phi) = A\omega^2\sin[\pi + (\omega t + \phi)]$$

These relationships indicate that the velocity vector leads the displacement vector by $\pi/2$ and the acceleration vector leads the displacement vector by π.



Energy in S.H.M.

During SHM, a particle has kinetic as well as potential energy.

(1) **Potential energy:** Potential energy depends on the displacement of the particle from its mean position.

Potential energy is due to the work done against the restoring force.

So, $U = -\int dw = -\int_0^y F dy = \int_0^y ky\, dy = \frac{1}{2}ky^2$

$\therefore$ Potential Energy, $U = \frac{1}{2}m\omega^2 y^2$ [As $\omega^2 = k/m$]

$U = \frac{1}{2}m\omega^2 a^2 \sin^2 \omega t$ [As $y = a\sin\omega t$]

Important points

(i) At extreme positions, potential energy is maximum and equal to total energy.

$U_{max} = \frac{1}{2}ka^2 = \frac{1}{2}m\omega^2 a^2$ when $y = \pm a$; $\omega t = \pi/2$; $t = T/4$

(ii) At the mean position, potential energy is minimum.

$U_{min} = 0$ when $y = 0$; $\omega t = 0$; $t = 0$

(2) **Kinetic energy:** Kinetic energy of the particle is due to the velocity during SHM.

Kinetic Energy, $K = \frac{1}{2}mv^2$

$K = \frac{1}{2}ma^2\omega^2 \cos^2 \omega t$ [As $v = a\omega\cos\omega t$]

$K = \frac{1}{2}m\omega^2(a^2 - y^2)$ [As $v = \omega\sqrt{a^2 - y^2}$]

Important points

(i) At the mean position, kinetic energy is maximum and equal to total energy at mean position.

$K_{max} = \frac{1}{2}m\omega^2 a^2$ when $y = 0$; $t = 0$; $\omega t = 0$

(ii) At the extreme position, kinetic energy is minimum.

$K_{min} = 0$ when $y = a$; $t = T/4$, $\omega t = \pi/2$

(3) **Total energy:** The total energy is the sum of the kinetic and potential energies.

Total mechanical energy = Kinetic energy + Potential energy

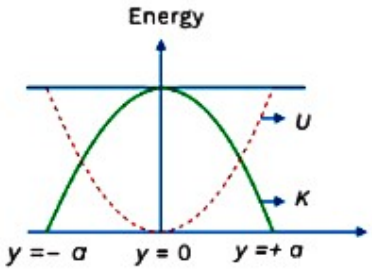
$E = \frac{1}{2}m\omega^2(a^2 - y^2) + \frac{1}{2}m\omega^2 y^2 = \frac{1}{2}m\omega^2 a^2$

Total energy always remains constant.

(4) **Energy position graph:** Kinetic energy (K) $= \frac{1}{2}m\omega^2(a^2 - y^2)$

Potential Energy (U) $= \frac{1}{2}m\omega^2y^2$

Total Energy (E) $= \frac{1}{2}m\omega^2a^2$



From the graph, we can see that,

(i) At mean position, kinetic energy is maximum and at extreme position, it is minimum.

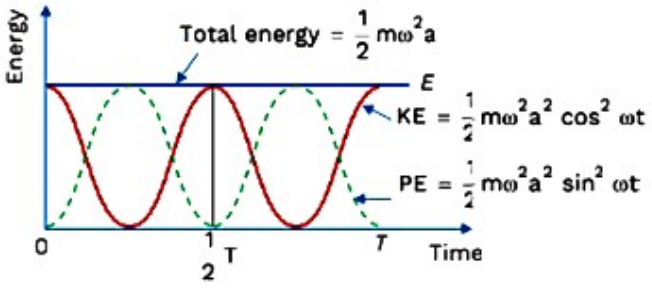
(ii) At mean position, potential energy is minimum and at extreme position, it is maximum.

(iii) Total mechanical energy remains constant.

(5) Kinetic Energy, $K = \frac{1}{2}m\omega^2a^2\cos^2\omega t = \frac{1}{4}m\omega^2a^2(1+\cos 2\omega t)$

Potential Energy, $U = \frac{1}{2}m\omega^2a^2\sin^2\omega t = \frac{1}{4}m\omega^2a^2(1-\cos 2\omega t)$

From the graph, it can be seen that the frequency of kinetic and potential energies vary with double frequency as that of the SHM frequency. The time period of the energies is half of the time period of SHM.



Q **For a particle executing SHM, the kinetic energy k is given by The equation of its displacement can be**

(a) $\left(\frac{k_0}{m\omega^2}\right)^{1/2}\sin\omega t$ **(b)** $\left(\frac{2k_0}{m\omega^2}\right)^{1/2}\sin\omega t$

(c) $\left(\frac{2\omega^2}{mk_0}\right)^{1/2}\sin\omega t$ **(d)** $\left(\frac{2k_0}{m\omega_0}\right)^{1/2}\sin\omega t$

Sol. (b)

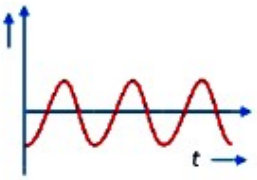
If m is the mass, r is the amplitude of oscillation, then maximum kinetic energy,

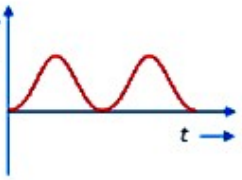
$$k_0 = \frac{1}{2}m\omega^2 r^2 \text{ or } r = \left(\frac{2k_0}{m\omega^2}\right)^{\frac{1}{2}}$$

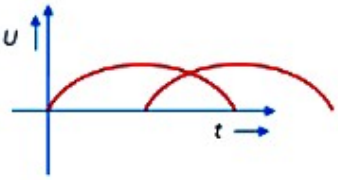
The displacement equation can be,

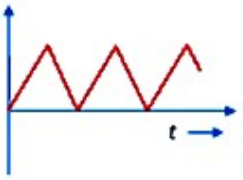
$$y = r\sin\omega t = \left(\frac{2k_0}{m\omega^2}\right)^{\frac{1}{2}}\sin\omega t$$

Q **As a body performs S.H.M., its potential energy varies with time as indicated in**

(a) 

(b) 

(c) 

(d) 

Sol. (b)

PE varies from zero to maximum. It is always positive sinusoidal function.

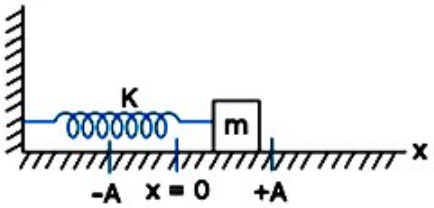
Systems Performing SHM

Systems Executing Simple Harmonic Motion

It is not physically possible to have absolute simple harmonic motion. This section describes some systems executing SHM.

Oscillations due to Spring

A spring block system is the simplest system shown in the figure below.



A spring having spring constant k is attached to a blok of mass m from one end and its other end is fixed to the wall. The block is on a frictionless surface.

Now, if the block is pushed or pulled by a small distance, then it performs SHM about x=0. -A and +A are the extreme displacements from the mean position.

At at any time instant, if x is the displacement from the mean position, then the restoring force F on the block is

$$F(x) = -kx \quad ...(1)$$

$$ma = -kx$$

Above equation is the SHM equation, so the system executes SHM. Then,

$$\omega = \sqrt{\frac{k}{m}}$$

And the period, T, for the oscillator is given by

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Q. A block of mass 5 kg is attached to a spring of k = 500 Nm^{-1}. It slides without friction over a horizontal surface. Now, it is displaced from equilibrium position by 10 cm and released. Find the time period of oscillation, the maximum speed and maximum acceleration during the oscillation.

Sol. Time period of oscillation, $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{5}{500}} = \frac{\pi}{5} = 0.628\,s$

Maximum speed, $v_{max} = A\omega = 0.1\sqrt{\frac{k}{m}} = 0.1\sqrt{\frac{500}{5}} = 0.1\times 10 = 1 ms^{-1}$

Maximum acceleration, $a_{max} = A\omega^2 = 0.1\times\frac{k}{m} = 0.1\times\frac{500}{5} = 0.1\times 100 = 10\ ms^{-2}$

Combinations of Spring

Series combination

For two springs with spring constant k_1 and k_2 be connected in series with a block of mass m, and x = displacement of the block from equilibrium position towards right (assuming positive direction) and x_1, x_2 as the extensions in the two springs, then

$x_1 = \frac{T}{k_1}$ and $x_2 = \frac{T}{k_2}$, where T = Restoring force in the spring.

Also, $x = x_1 + x_2$

$$\Rightarrow x = \frac{T}{k_1} + \frac{T}{k_2} = T\left(\frac{1}{k_1} + \frac{1}{k_2}\right)$$

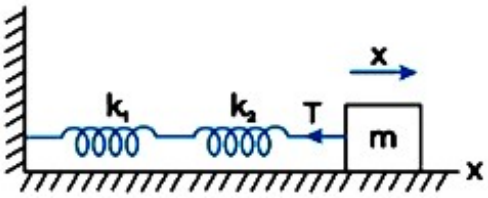
$$\Rightarrow T = \frac{k_1k_2}{k_1 + k_2}x$$

Net force on the block = T, so by Newton's 2nd law,

$$m\frac{d^2x}{dt^2} = -T$$

$$\Rightarrow m\frac{d^2x}{dt^2} = -\left(\frac{k_1k_2}{k_1 + k_2}\right)x$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k_1k_2}{m(k_1 + k_2)}x$$



On comparing with standard equation of SHM, $\frac{d^2x}{dt^2} = -\omega^2x$,

$$\omega^2 = \frac{k_1k_2}{m(k_1 + k_2)}$$

So, $\omega = \sqrt{\frac{k_1 k_2}{m(k_1 + k_2)}} = \sqrt{\frac{k_{eq}}{m}}$

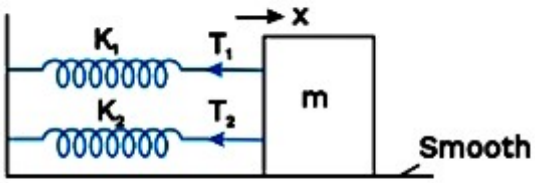
or $T = 2\pi \sqrt{\frac{m}{\sqrt{\left(\frac{1}{k_1} + \frac{1}{k_2}\right)}}}$

So, equivalent spring constant for springs connected in series is given by,

$$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} \ldots.$$

Parallel combination

Consider that two springs having spring constants k_1 and k_2 are connected to a block of mass m as shown.



For a displacement x for the block,

$T_1 = -k_1 x$

$T_2 = -k_2 x$

Net force on the block is $T_1 + T_2$. So, by Newton's 2nd law,

$$m\frac{d^2x}{dt^2} = -(T_1 + T_2)$$

$$\Rightarrow m\frac{d^2x}{dt^2} = -(k_1 + k_2)x$$

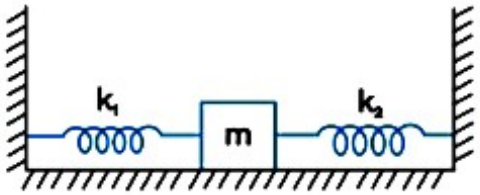
$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{(k_1 + k_2)}{m}x$$

Comparing with $\frac{d^2x}{dt^2} = -\omega^2 x$, we have

$$\omega = \sqrt{\frac{k_1 + k_2}{m}}$$

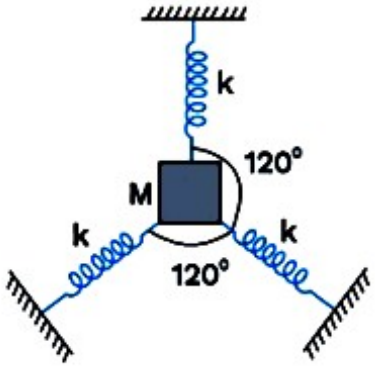
$$\Rightarrow T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$$

Similarly, the time period of the system shown below is also $T = 2\pi\sqrt{\frac{m}{k_1 + k_2}}$ as the system shown below also contains springs in parallel.



Equivalent spring constant for springs in parallel given as, $k_{eq} = k_1 + k_2$

Q. **Find the time period of oscillation for the block shown in the figure below.**

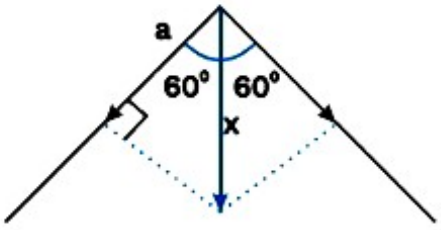


Sol. On displacing the block by x in downward direction, the upper spring elongates by x. If there is a compression "a" in the other two springs, then

$a = x \cos 60°$

$$a = \frac{x}{2}$$

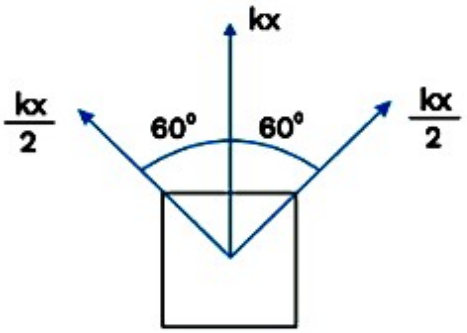
Net force on the block $= -\left(kx + \frac{kx}{2}\right)$



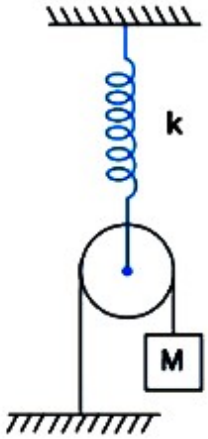
By Newton's 2nd law, $m\frac{d^2x}{dt^2} = -\frac{3}{2}kx$

$$\Rightarrow \omega^2 = \frac{3k}{2M}$$

And, time period, $T = 2\pi\sqrt{\frac{2M}{3k}}$.



Q. **For the ideal pulley arrangement, find the time period of oscillation of the block.**



Sol. Let x_0 be the extension in the spring at initial equilibrium, then tension in spring is kx_0.

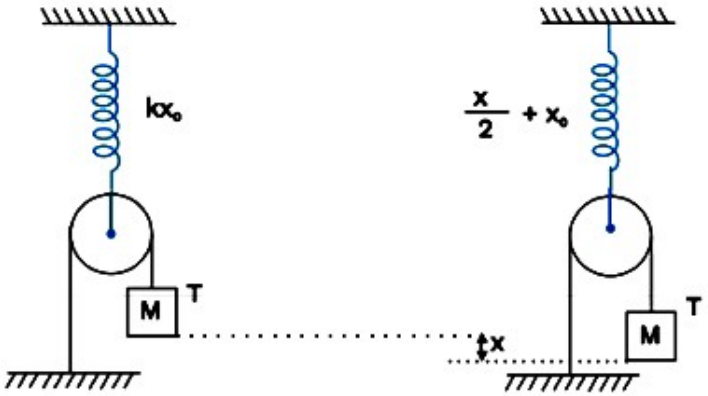
Then, tension in string to which is block is attached is T.

$$2T = kx_0 \text{ and } T = Mg$$

On moving the block down by a distance x, then

additional extension in spring = $\frac{x}{2}$.

Then, spring force = $k\left(x_0 + \frac{x}{2}\right)$.



Equilibrium Condition Displaced Block

Tension is string, $T' = \frac{k}{2}\left(x_0 + \frac{x}{2}\right)$

Net force on the block

$$F = Mg - \frac{k}{2}\left(x_0 + \frac{x}{2}\right)$$

$$= Mg - \frac{kx_0}{2} - \frac{kx}{4} \qquad [\because Mg = T]$$

$$= T - \frac{kx_0}{2} - \frac{kx}{4} \qquad \left[\because T = \frac{kx_0}{2}\right]$$

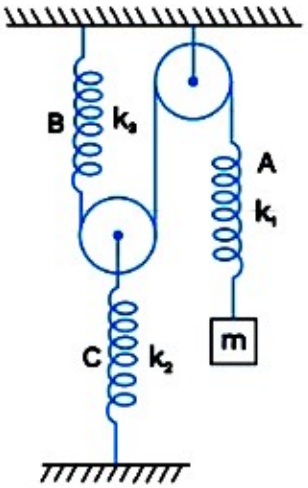
$$\Rightarrow F = -\frac{kx}{4}$$

Then, effective spring constant = $\frac{k}{4}$.

Time period of oscillation = $2\pi\sqrt{\frac{M}{k_{eq}}}$

$$= 2\pi\sqrt{\frac{4M}{k}}$$

Q. **Find the time period of oscillation in following arrangement.**



Sol. Displace the block of mass M in downward direction by x.

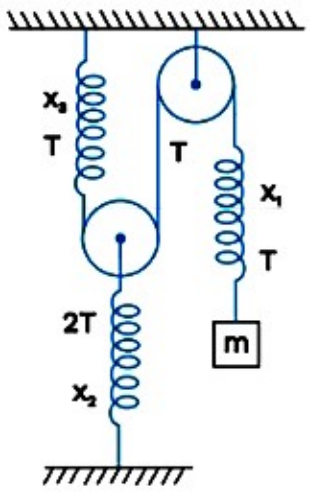
Assume elongation in the springs A, B and C are x_1, x_2 and x_3 respectively.
By constraint relations,

$$x = x_1 + 2x_2 + x_3$$

If T is tension in spring A and B respectively then tension in spring C will be 2T.

$$T = k_1x_1 = k_3x_3$$

$$2T = k_2x_2$$



Putting the value of x_1, x_2 are x_3,

$$x = \frac{T}{k_1} + \frac{4T}{k_2} + \frac{T}{k_3}$$

$$\text{or } T = \frac{x}{\left(\frac{1}{k_1} + \frac{4}{k_2} + \frac{1}{k_3}\right)}$$

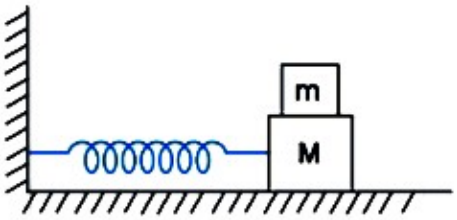
So, effective spring constant is,

$$k_{eff} = \frac{1}{\left(\frac{1}{k_1} + \frac{4}{k_2} + \frac{1}{k_3}\right)}$$

Then, time period of oscillation is

$$T = 2\pi\sqrt{\frac{m}{k_{eff}}} = 2\pi\sqrt{m\left(\frac{1}{k_1}+\frac{4}{k_2}+\frac{1}{k_3}\right)}$$

Q. **A block of mass M is executing SHM with frequency f and amplitude A. Now, when the block of mass M passes through the equilibrium position, a small block of mass m is kept on block of mass M and the two blocks move together. Find the new frequency and the new amplitude of oscillation.**



Sol. For the oscillation of the block of mass M,

$$\therefore f = \frac{1}{2\pi}\sqrt{\frac{k}{M}}$$

If f' is the new frequency and A' is the new amplitude of oscillation after adding the block of mass m.

Now, total mass = M + m.

$$\therefore f' = \frac{1}{2\pi}\sqrt{\frac{k}{(m+M)}}$$

$$\text{Now, } \frac{f'}{f} = \sqrt{\frac{M}{(m+M)}}$$

$$\therefore f' = f\sqrt{\frac{M}{(m+M)}} \qquad \ldots\ldots (1)$$

So, f' < f.

When mass M passes through its mean position, it has maximum speed.

Let v and V be the speed of block of mass M and M + m respectively at equilibrium position.

By linear momentum conservation,

$Mv = (m + M)\, V$

Also, at equilibrium position

$v = A\omega = A \times 2\pi f$

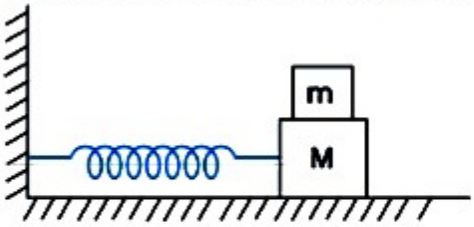
$\therefore MA(2\pi f) = (m + M)A'(2\pi f')$

$$\Rightarrow \frac{A'}{A} = \frac{M}{(m+M)} \times \frac{f}{f'}$$

Using equation (1), we have,

$$\text{New amplitude, } A' = A\sqrt{\frac{M}{(m+M)}}$$

Q. **A block of mass m attached to the free end of a spring of force constant k is kept on a smooth horizontal table as shown in figure. A block having mass m is kept on the top rough surface of the first block. The coefficient of static friction between the surfaces of two blocks is μ. If we assume that two bodies move together as a single unit, then find the time period of oscillation of the system and the maximum oscillation amplitude that allows the two bodies to move as a single unit.**



Sol. On displacing the block of mass M from its equilibrium position and releasing, it performs SHM

If x is the displacement of block at any instant, then the equation of SHM is

$$(m + M)\frac{d^2x}{dt^2} = -kx$$

$$\text{or } \frac{d^2x}{dt^2} = -\frac{k}{(m+M)}x$$

or $\frac{d^2x}{dt^2} = -\omega^2 x$

where $\omega = \sqrt{\frac{k}{(m+M)}}$

$\therefore$ Time period of SHM $= \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{(M+m)}{k}}$

Now, for the two blocks to move as a single unit, at the extreme position, the force on m should be less than maximum static friction force.
If A is the amplitude of SHM,
then, for the two bodies to move as a single unit,

$m\omega^2 A \leq \mu m g$

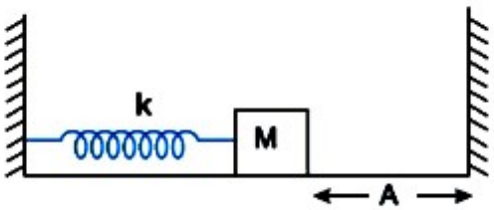
or $A \leq \frac{\mu g}{\omega^2}$

$\therefore A \leq \frac{\mu g(m+M)}{k}$

Thus, the maximum amplitude of oscillations that permits the two bodies to move as a single unit is

$A = \frac{\mu g(m+M)}{k}$

Q. **In the system given below, the spring is compressed by a length 2A and released. When the block collides with the wall, it loses two third of its kinetic energy. After what time the spring has maximum compression after releasing it.**

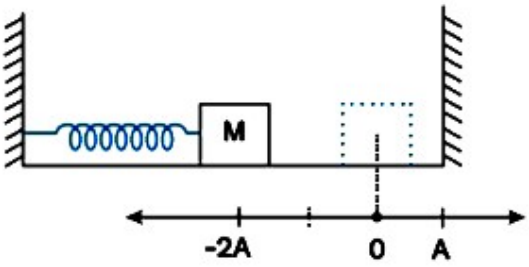


Sol. On releasing the block from -2A, initial velocity of block is zero.

Total energy = Potential energy + kinetic energy

$$= \frac{1}{2}k(2A^2) + 0$$

$$= 2kA^2.$$



Now, kinetic energy of the block at the instant it strikes the wall

$$= 2kA^2 - \frac{1}{2}kA^2$$

$$= \frac{3}{2}kA^2$$

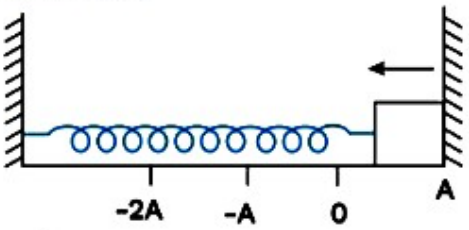
Before the collision, the motion of the block will be SHM of amplitude 2A and time period $2\pi\sqrt{\frac{m}{k}}$.

During motion of the block from —2A to A, it has covered a phase of $\left(\frac{2\pi}{3}\right)$.

So, the time taken by the block in moving from —2A to A,

$$t_1 = \frac{T}{2\pi} \times (\text{Phase interval})$$

$$= \frac{T}{2\pi} \times \left(\frac{2\pi}{3}\right) = \frac{T}{3}$$



Just after the collision, kinetic energy of the block $\frac{1}{2}kA^2$ (One third of the kinetic energy before collision)

Now the motion of block will be SHM of lesser amplitude. If new amplitude is A' , then

$$\text{K.E.} = \frac{1}{2}k(A'^2 - x^2)$$

$$\Rightarrow \frac{1}{2}kA^2 = \frac{1}{2}k(A'^2 - A^2)$$

$$\Rightarrow A'^2 = 2A^2$$

or $A' = A\sqrt{2}$

Let the equation of displacement of the block after collision is (Taking collision instant t = 0)

$$x = A\sqrt{2}\sin(\omega t + \phi)$$

At $t = 0$, $x = A$ so, $\phi = \frac{\pi}{4}$ or $\frac{3\pi}{4}$

As velocity is negative, so, $\phi = \frac{3\pi}{4}$

Compression in spring will be maximum when the block is at $x = -A\sqrt{2}$

$$\sin\left(\omega t + \frac{\pi}{4}\right) = -1$$

$$\Rightarrow \quad \omega t + \frac{3\pi}{4} = \frac{3\pi}{2}$$

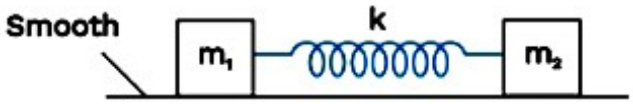
$$\Rightarrow \frac{2\pi}{T}t = \frac{3\pi}{4}$$

$$\Rightarrow t = \frac{3\pi}{8}$$

So, the total time $= \frac{T}{3} + \frac{3T}{8} = \frac{17T}{24} = \frac{17\pi}{12}\sqrt{\frac{M}{k}}$

Two Block Oscillations

Consider two blocks as shown in the figure below. On slightly extending and leaving the system, it oscillates.



At any instant, if x= extension in spring and T=tension in spring,

Acceleration of block of mass m_1, $a = \frac{T}{m_1}$ (towards right)

From the reference frame of m_1.

Net force on $m_2 = T + m_2 a$

$$\Rightarrow m_2\frac{d^2x}{dt^2} = -(T + m_2 a)$$

$$\Rightarrow m_2\frac{d^2x}{dt^2} = -\left[kx + \frac{m_2 k x}{m_1}\right]$$

$$\Rightarrow \frac{d^2x}{dt^2} = -k\left[\frac{m_1 + m_2}{m_1 m_2}\right]x$$

So, $\omega^2 = k\left[\frac{m_1 + m_2}{m_1 m_2}\right]$

$$\Rightarrow T = 2\pi\sqrt{\frac{\left(\frac{m_1 m_2}{m_1 + m_2}\right)}{k}}$$

Here, $\frac{m_1 m_2}{m_1 + m_2} = \mu$ is called reduced mass.

We can analyse the oscillation of the system from its centre of mass frame.

As there is no external force, the centre of mass of the system remains at rest.

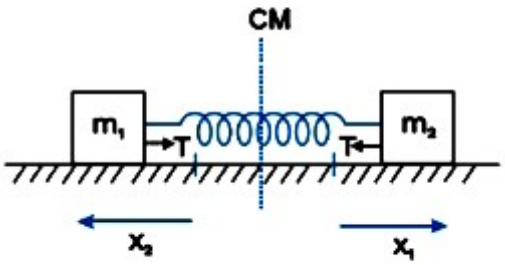
If x_1 and x_2 are the displacement of m_1 and m_2 with respect to centre of mass and x is the total extension in the spring. Then,

$$x_1 + x_2 = x$$

Differentiating above equation wrt time,

$$\frac{dx_1}{dt} + \frac{dx_2}{dt} = \frac{dx}{dt}$$

Again, differentiating wrt time,



$$\frac{d^2x_1}{dt^2} + \frac{d^2x_2}{dt^2} = \frac{d^2x}{dt^2}$$

$$\Rightarrow \frac{T}{m_1} + \frac{T}{m_2} = \frac{d^2x}{dt^2} \qquad \left(\because T = m_1\frac{d^2x_1}{dt^2} \;; T = m_2\frac{d^2x_2}{dt^2}\right)$$

$$\Rightarrow T\left(\frac{1}{m_1} + \frac{1}{m_2}\right) = \frac{d^2x}{dt^2}$$

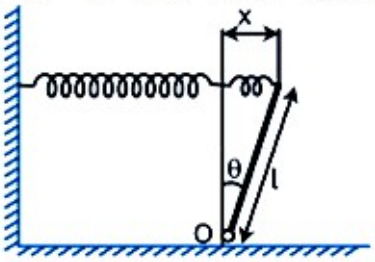
$$\Rightarrow -kx\left(\frac{m_1 + m_2}{m_1 m_2}\right) = \frac{d^2x}{dt^2} \qquad (\because T = -kx)$$

$$\Rightarrow \frac{d^2x}{dt^2} = \frac{-k(m_1 + m_2)}{m_1 m_2} x$$

$$\text{So, } \omega^2 = \frac{k(m_1 + m_2)}{m_1 m_2}$$

$$\Rightarrow T = 2\pi\sqrt{\frac{m_1 m_2}{k(m_1 + m_2)}}$$

Q. **A straight rod having mass m and length l is attached to a spring as shown. On rotating the rod slightly about the hinge, find the time period of oscillation of the rod. Ignore the effect of gravity.**



Sol. Consider the rod is displaced from its equilibrium position by a distance x,

Then there will be a restoring torque developed due to the spring i.e.,

$\tau = Fl$

Now, $F = -kx$

$\therefore \tau = -kxl$

As x is very small,

$$\therefore \frac{x}{l} = \theta \text{ or } x = l\theta \qquad \therefore \qquad \tau = -kl^2\theta \qquad \ldots(i)$$

Now, by Newton's second law for rotational motion,

$$\tau = l\alpha \qquad \text{where, } I = \frac{1}{3}ml^2, \alpha = \frac{d^2\theta}{dt^2}$$

$$\therefore \tau = \frac{1}{3}ml^2\frac{d^2\theta}{dt^2} \qquad \ldots(ii)$$

From (i) & (ii),

$$\frac{1}{3}ml^2\frac{d^2\theta}{dt^2} = -kl^2\theta$$

$$\therefore \quad \frac{d^2\theta}{dt^2} = -\frac{3k}{m}\theta$$

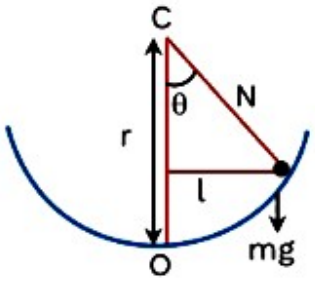
$$\text{or} \quad \frac{d^2\theta}{dt^2} = -\omega^2\theta \text{ with } \omega = \sqrt{\frac{3k}{m}}$$

This is standard equation of angular SHM

$$\therefore \text{ Time period } = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{3k}}$$

Q. **A small ball having a mass m is kept inside a concave surface. If the ball is displaced at a small distance from the centre of a smooth concave surface of radius r and released, then find the time period of oscillation of ball.**

Sol. On displacing the ball by a small distance from the centre O of the concave surface and releasing it, it oscillates about point O.



Component of weight of the ball is the restoring force on the ball.

$\therefore$ Restoring torque acting on ball due to the force of gravity $\tau = -m\,g\,l$

As θ is very small,

$\therefore$ $l = r\sin\theta = r\,\theta$

$\because$ $\tau = -mgr\theta$ ...(i)

As restoring torque $\tau \propto \theta$ and opposite to displacement,

Therefore, motion will be SHM

Now, $\tau = I\,\alpha$

where I = moment of inertia = mr^2

$$\alpha = \frac{d^2\theta}{dt^2}$$

$$\therefore \quad \tau = mr^2\frac{d^2\theta}{dt^2} \qquad \text{...(ii)}$$

From equation (i) and (ii), we get

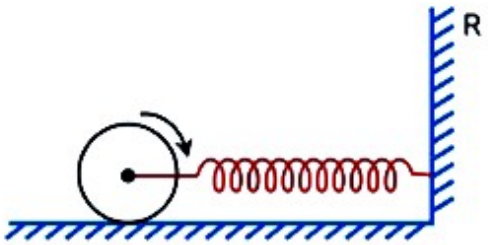
$$mr^2\frac{d^2\theta}{dt^2} = -mgr\theta$$

$$\therefore \quad \frac{d^2\theta}{dt^2} = -\frac{g}{r}\theta$$

or $$\frac{d^2\theta}{dt^2} = -\omega^2\theta \text{ with } \omega^2 = \frac{g}{r}$$

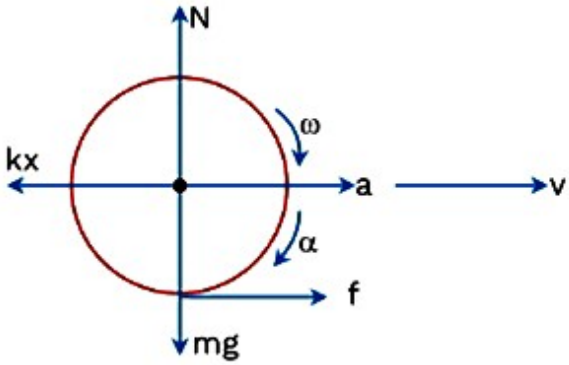
Therefore, time period of oscillations $= \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r}{g}}$

Q. **A solid sphere having mass m and radius R is kept on a rough surface as shown. An impulse is given to the sphere at its mean position and the sphere starts oscillating. Find the time period of oscillation, if there is no slipping anywhere.**



Sol. **Method - 1 (Force Method) :**

Draw the free body diagram of the sphere at a displacement x (towards right) from mean position while sphere is moving in right direction.



Here,

ω = angular velocity about centre,

v = linear velocity of centre,

α = angular acceleration about centre,

f = force of friction

By 2nd law of motion

$\Sigma F = ma \Rightarrow \quad f - kx = ma$ ...(i)

$\Sigma\tau = I\alpha$

$$\Rightarrow -fR = \frac{2}{5}mR^2\alpha$$

$$\Rightarrow f = -\frac{2}{5}mR\alpha \qquad \text{...(ii)}$$

For no slipping

$$a = R\alpha \qquad \text{...(iii)}$$

From (i), (ii), & (iii)

$$-\frac{2}{5}ma - kx = ma$$

$$\Rightarrow a = -\left(\frac{5k}{7m}\right)x = -\omega^2 x$$

i.e., $a \propto (-x)$

So, the motion is S.H.M. with time period, $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{7m}{5k}}$

Method-2 (Energy Method)

Since, there is no dissipative force acting on the spring mass system, hence total mechanical energy of the system remains conserved i.e.,

$$E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\omega^2 = \text{ constant}$$

As total energy is constant, so $\frac{dE}{dt} = 0$.

For no slipping, $v = \omega r$

$$\Rightarrow E = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mR^2\right)\left(\frac{v}{R}\right)^2$$

Differentiating the above equation w.r.t. time (t) and putting it equal to zero, we get

$$\frac{dE}{dt} = \frac{1}{2}k\left(2\times\frac{dx}{dt}\right) + \frac{1}{2}m\left(2v\frac{dv}{dt}\right) + \frac{1}{2}\times\frac{2}{5}m\times\left(2v\frac{dv}{dt}\right) = 0$$

As $v = \frac{dx}{dt}$, so v cancels out in the whole equation and we get,

$$\frac{dv}{dt} = -\left(\frac{5k}{7m}\right)x$$

i.e., $$a = -\left(\frac{5k}{7m}\right)x = -\omega^2 x$$

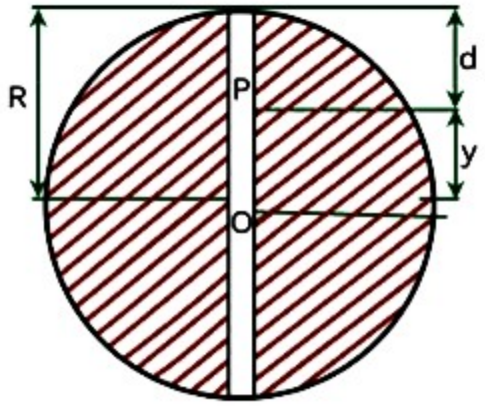
So, the motion is S.H.M. with $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{7m}{5k}}$

Q. **What is the oscillation time period of a body of mass m dropped in a tunnel along the diameter (2R) of the earth and also find the effect on the time period, if the tunnel is along a chord not along diameter of earth.**

Sol. (a) For the tunnel across the diameter, let the body is at point P as shown.

$\therefore$ If g' = acceleration due to gravity at P, then

$$g' = -g\left[1 - \frac{d}{R}\right] = -g\left[\frac{R-d}{R}\right]$$



The –ve sign shows that as d increases, g' decreases.

Let y be the distance of body from the earth's centre.

$\therefore$ $R - d = y$

$\therefore$ $g' = -\frac{g}{R}y$

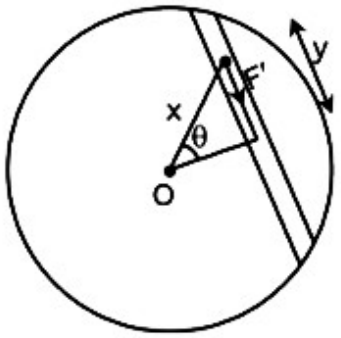
As $\frac{g}{R}$ = constant, so above equation is similar to SHM equation.

$g' = -\omega^2 y$, where $\omega^2 = \frac{g}{R}$

$\Rightarrow$ Time period, $T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{6.4\times10^6}{9.8}} = 5077.6\,s = 84.62$ minutes

Time period (T) does not depend on mass of the body and depends on R and g (both are constant).

(b)



For a tunnel along a chord and body at a distance x from the centre of earth at any time, if F′ is the restoring force acting on the body due to gravity, then

$$F' = F\sin\theta, \quad \text{where } F = -\frac{GMm}{R^3}x$$

$$\therefore \qquad F' = -\left(\frac{GMm}{R^3}\right)x\left[\frac{y}{x}\right] = -\frac{GMm}{R^3}\cdot y$$

or $F' = -ky$

$\therefore$ Motion is again SHM.

$$\therefore \qquad T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{mR^3}{GMm}} = 2\pi\sqrt{\frac{R^3}{GM}}$$

Also, $g = \dfrac{GM}{R^2}$

$$\therefore \qquad T = 2\pi\sqrt{\frac{R}{g}} = 84.62 \text{ minute}$$

$\therefore$ Motion is SHM with same time period for both the tunnels across the diameter and the across any chord.

Simple Pendulum

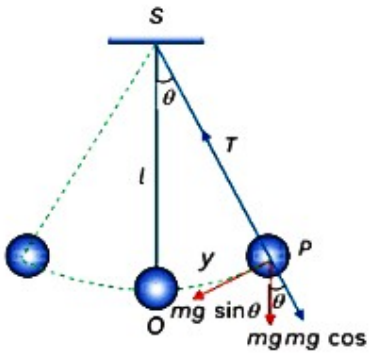
In an ideal pendulum, there is a mass (bob) attached at one end of an inextensible string. The mass can have angular displacements as shown.

For m = mass of bob

l =Length of string

On displacing the bob by a small angle θ from the vertical,

Restoring force on the bob,



$F = -mg\sin\theta$

or $F = -mg\theta$ (When θ is small $\sin\theta \simeq \theta = \frac{\text{Arc}}{\text{Length}} = \frac{OP}{l} = \frac{x}{l}$)

or $F = -mg\frac{x}{l}$

$\therefore \quad a = -\left(\frac{g}{l}\right)x = -\omega^2 x$ (Spring factor)

So, time period, $\omega = \sqrt{\frac{g}{l}} = \frac{2\pi}{T} \Rightarrow T = 2\pi\sqrt{\frac{l}{g}}$

Important points

(i) Time period of simple pendulum does not depend upon the amplitude of SHM. However, for larger angular amplitudes, the motion of the pendulum bob is not SHM, it is considered oscillatory.

(ii) Time period of simple pendulum does not depend on the mass of the bob.

(iii) Time period $T \propto \sqrt{l}$ where l is the effective length of the pendulum and it is the distance between centre of mass of bob and point of suspension.

(iv) For the pendulum length comparable to Earth's radius, $T = 2\pi\sqrt{\frac{1}{g\left[\frac{1}{l}+\frac{1}{R}\right]}}$

(a) If $l << R$, then $\frac{1}{l} >> \frac{1}{R}$ so $T = 2\pi\sqrt{\frac{l}{g}}$

(b) If $l >> R(\to \infty)$ so $T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{6.4\times10^6}{10}} \cong 84.6$ minutes

(c) If $l = R$ so $T = 2\pi\sqrt{\frac{R}{2g}} \cong 1$ hour

(v) **Pendulum in a lift :**

(a) For the lift moving with constant velocity,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

(b) For the lift accelerating upwards,

$$T = 2\pi\sqrt{\frac{l}{g+a}}$$

(c) For the lift accelerating downwards,

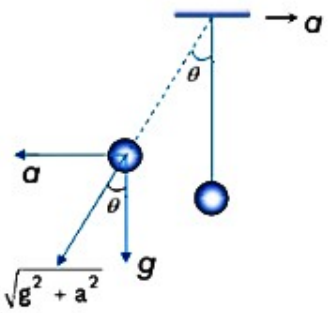
$$T = 2\pi\sqrt{\frac{l}{g-a}}$$

(d) If the lift is moving downward with acceleration $a = g$

$$T = 2\pi\sqrt{\frac{l}{g-g}} = \infty$$

It shows that there will be no oscillation in a pendulum under free fall.

(vi) For a pendulum whose point of suspension is accelerating horizontally with acceleration a.



$$T = 2\pi\sqrt{\frac{l}{(g^2+a^2)^{1/2}}} \text{ and } \theta = \tan^{-1}(a/g)$$

(vii) For a simple pendulum suspended in a vehicle moving with constant speed v in a circle of radius r.

$$T = 2\pi\frac{\sqrt{l}}{\sqrt{g^2+\left(\frac{v^2}{r}\right)^2}}$$

(viii) **Second's Pendulum:** It is a pendulum with a time period of two seconds.

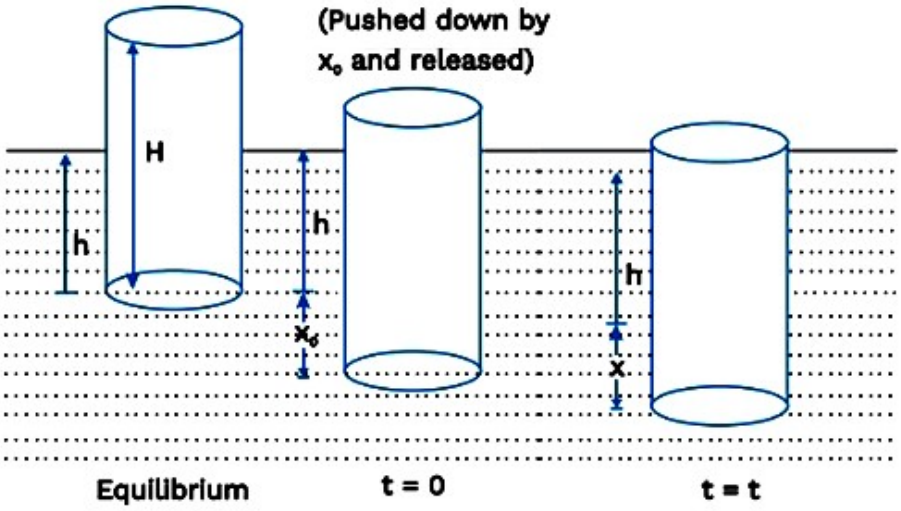
On putting $T = 2$ sec and $g = 9.8 m/sec^2$ in $T = 2\pi\sqrt{\frac{l}{g}}$,

$$l = \frac{4 \times 9.8}{4\pi^2} = 0.993\,m = 99.3\ cm$$

So, the second's pendulum length = 99.3 cm or nearly 1 meter on earth surface.

Q. **Find the time period of the SHM of a solid cylinder of density ρ height h which is gently pushed down and left while it was floating in a non viscous liquid of density ρ_0.**

Sol. Consider that at equlibrium position of the cylinder, a length h of the cylinder is inside the liquid.



At initial equilbrium,

$AH\rho g = Ah\rho_0 g$ [Net force on the cylinder is zero, so buoyant force= Weight]

$\Rightarrow H\rho = h\rho_0$

Now if at any instnat t, the cylinder is displaced by x in downward direction (i.e., positive) direction.

By Newton's second law,

$$M\frac{d^2x}{dt^2} = -\left[A(h+x)\rho_0 g - AH\rho g\right]$$

$$= -\left[A\rho_0 gh + A\rho_0 gx - A\rho gH\right]$$

$$(\rho AH)\frac{d^2x}{dt^2} = -(A\rho_0 g)\, x$$

$$\frac{d^2x}{dt^2} = -\left(\frac{\rho_0 g}{\rho H}\right)x$$

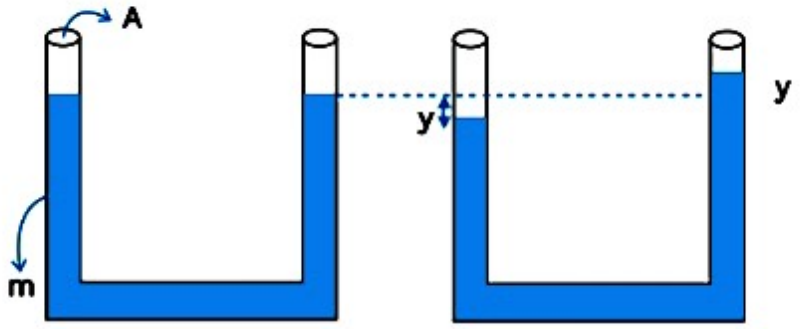
Comparing with $\frac{d^2x}{dt^2} = -\omega^2 x$

$$\omega^2 = \frac{\rho_0 g}{\rho H}$$

$$\boxed{T = 2\pi\sqrt{\frac{\rho H}{\rho_0 g}}}$$

Oscillations of Liquid in a U-tube

On displacing a non-viscous liquid slightly from its equilibrium position in a U-tube, it performs oscillations.



Consider that an instant, in the left arm the liquid is y distance downwards from mean position, then it will move in upward direction by same amount in the right arm.

The difference in the length of liquid column in two arms is 2y i.e., unbalanced head of liquid is 2y.

This head applies force on rest of the liquid and acts as a restoring force on rest of the liquid.

F = — Force applied by liquid column of height 2y

= — [Pressure at a depth 2y] [A]

= — [2y ρg] × A

= —2y ρ g A

Now acceleration of the liquid column

$$a = \frac{F}{m}$$

or $a = -\left(\frac{2\rho g A}{m}\right) y$ ..... (i)

or $a \propto (-y)$

therefore, oscillations of liquid column are simple harmonic.

Comparing with $a = -\omega^2 y$

$$\omega^2 = \frac{2\rho g A}{m}$$

or $$\omega = \sqrt{\frac{2\rho g A}{m}}$$

or $$T = \frac{2\pi}{\omega}$$

$$\boxed{T = 2\pi\sqrt{\frac{m}{2\rho g A}}}$$

Again, length of the liquid column in U tube is l, then

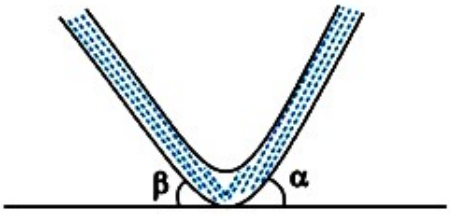
$M = Al\rho$

Then, $$T = 2\pi\sqrt{\frac{Al\rho}{2\rho g A}}$$

or $$\boxed{T = 2\pi\sqrt{\frac{l}{2g}}}$$

NOTE:

For the liquid shown in the figure below, the time period of oscillation of liquid column having length l is given by:

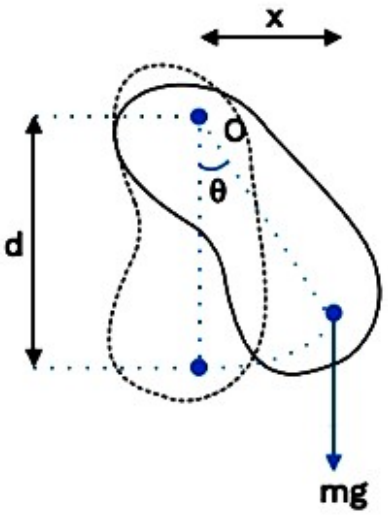


$$\boxed{T = 2\pi\sqrt{\frac{l}{g(\sin\alpha + \sin\beta)}}}$$

Physical Pendulum

Any rigid body hinged from a fixed point is termed as a physical pendulum if it executes oscillations about that fixed point.

Consider a body hinged at O and free to rotate in the vertical plane. Let mass of the body be m and d be the distance of its centre of mass from point O.



For small angular displacement θ, restoring torque,

$\tau = -mg\,x$

$= -mg\,d\sin\theta \quad [x = d\sin\theta]$

$\tau = mgd(-\theta) \qquad$ (for small θ, $\sin\theta = \theta$)

$\Rightarrow \tau \propto (-\theta)$, which is the SHM equation.

Angular acceleration, $\alpha = -\dfrac{mgd}{I_o}\theta$, where I_o is the moment of inertia about the point of pivot O.

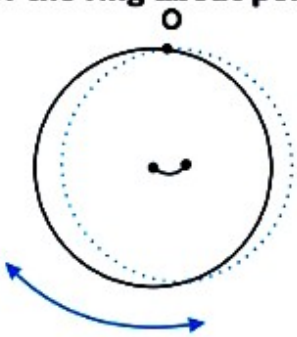
On comparing with $\alpha = -\omega^2\theta$, we have

$$\omega^2 = \frac{mgd}{I_o}$$

or $$\omega = \sqrt{\frac{mgd}{I_o}}$$

or $$T = 2\pi\sqrt{\frac{I_o}{mgd}}$$

Q. **A ring of mass M is suspended at a point on the periphery O as shown. Find the time period of oscillation of the ring about point O.**



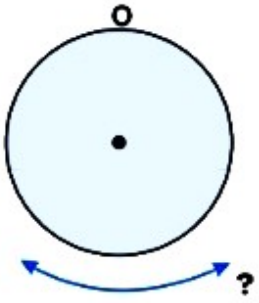
Sol. Moment of inertia of the ring abut horizontal axis passing through O.

$$I_o = MR^2 + MR^2 = 2\,MR^2$$

Now time period of small oscillation,

$$T = 2\pi\sqrt{\frac{I_o}{Mgd}} = 2\pi\sqrt{\frac{2MR^2}{MgR}}. \text{ Or } T = 2\pi\sqrt{\frac{2R}{g}}$$

Q. **A disc of mass M is suspended at O. Find the time period of oscillation of the disc about point O.**



Sol. Moment of inertia of the disc about horizontal axis passing through O.

$$I = \frac{MR^2}{2} + MR^2 = \frac{3}{2}MR^2$$

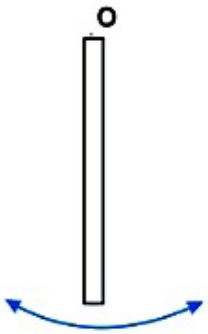
Distance of centre of mass from O, d = R

So, time period of small oscillations,

$$T = 2\pi\sqrt{\frac{I}{Mgd}} = 2\pi\sqrt{\frac{\frac{3}{2}MR^2}{MgR}}$$

or $T = 2\pi\sqrt{\frac{3R}{2g}}$

Q. **What is the oscillation time period of a rod of mass M and length L hinged from its one end in the vertical plane?**



Sol. Moment of inertia of the rod about a horizontal axis passing through point O,

$$I_o = \frac{ML^2}{3}$$

Distance of centre of mass from O, $d = \frac{L}{2}$

So, time period of small oscillation,

$$T = 2\pi\sqrt{\frac{I_o}{Mgd}} = 2\pi\sqrt{\frac{\left(\frac{ML^2}{3}\right)}{Mg\left(\frac{L}{2}\right)}}$$

$$T = 2\pi\sqrt{\frac{2L}{3g}}$$

Q. **A rod is hinged by a nail at a point 0.5 m from its one end. The rod has a length of 2 m and its mass is 2 kg. Find the time period of oscillations of the rod?**

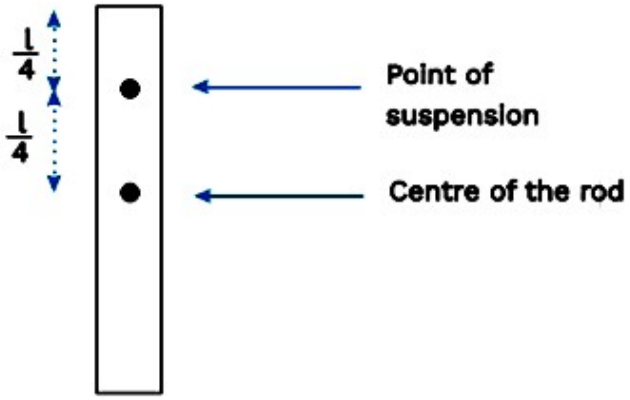
Sol. Mass of rod = m = 2 kg

Length of rod = l = 2m

Moment of inertia about the hinge point,

$$I = I_{CM} + md^2 = \frac{ml^2}{12} + m\left(\frac{l}{4}\right)^2 = \frac{7ml^2}{48}$$

And the distance of centre of mass of rod from the point of suspension is $d = \frac{l}{4}$ as shown in figure.



The time period of the physical pendulum for small oscillation is, $T = 2\pi\sqrt{\frac{I_o}{mgd}}$

$$T = 2\pi\sqrt{\frac{7ml^2}{48 \times mg \times \frac{l}{4}}}$$

$$= 2\pi\sqrt{\frac{7l}{12g}} = 2\pi\sqrt{\frac{7 \times 2}{12 \times 10}} = \pi\sqrt{\frac{7}{15}}s = 2.146\ s$$

Q. **If the length of simple pendulum of a clock increases by 2%, how much loss or gain of seconds per day will be observed?**

Sol. As the percentage change in length is very less, so formulae of error analysis are applicable here.

So, $\frac{\Delta T}{T} = \frac{1}{2}\frac{\Delta L}{L}$

Here, T is time in second for the whole day = 86400 s

Given $\frac{\Delta L}{L} = 2\% = \frac{2}{100}$

$\therefore \quad \frac{\Delta T}{86400} = \frac{1}{2} \times \frac{2}{100}$

$\therefore \Delta T = 864$ s

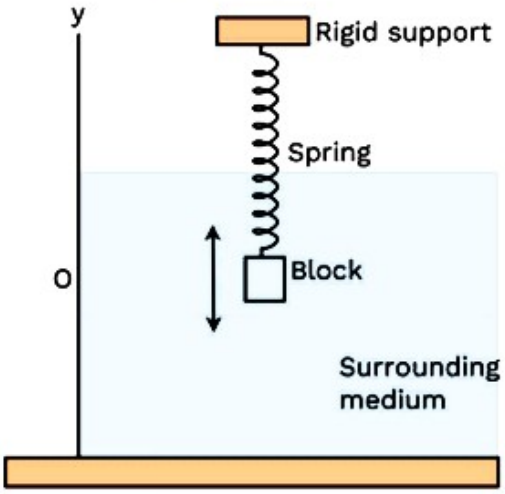
As length is increasing, the time period T will become large and the clock will go slow, so it will lose 864 s.

Damped and Forced Oscillations

DAMPED SIMPLE HARMONIC MOTION

In the section of systems performing SHM, we stated that absolute simple harmonic motion is not possible practically. It is so because air drag and friction present in natural systems act as resistance and the oscillations die out after some time as these resistances dissipate the energy gradually. The system is said to execute **damped oscillations**. The energy of the system gets dissipated continuously but the oscillations are approximately periodic for small damping.

Consider a spring-block system shown in the figure.



On pulling down and releasing the block, angular frequency of oscillation will be,

$$\omega = \sqrt{\frac{k}{m}}.$$

The surrounding medium is air which exerts a damping force on the motion of the block, and the mechanical energy of the spring-block system will decrease. This loss of energy appears in the form of heat in the surrounding medium and the block. Damping force depend upon the surrounding medium. It is more for liquids.

If the damping force is denoted by $\vec{F}_d$, then

$$\vec{F}_d = -b\vec{v} \quad \text{where b=damping coefficient}$$

Value of b depends on characteristics of the medium (i.e., viscosity) and size and shape of the block, etc.

Total force on the mass at any time, $\vec{F} = -k\vec{x} - b\vec{v}$

If $\bar{a}(t)$ is the acceleration of the mass at time t, then by Newton's 2nd law,

$ma(t) = -kx(t) - bv(t)$

For one dimensional motion, we can write,

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

The solution of above differential equation gives the motion of the block under the influence of a damping force which is proportional to velocity.

The solution is of the form of

$$x(t) = Ae^{\frac{-bt}{2m}}\cos\left(\omega_d t + \phi\right)$$

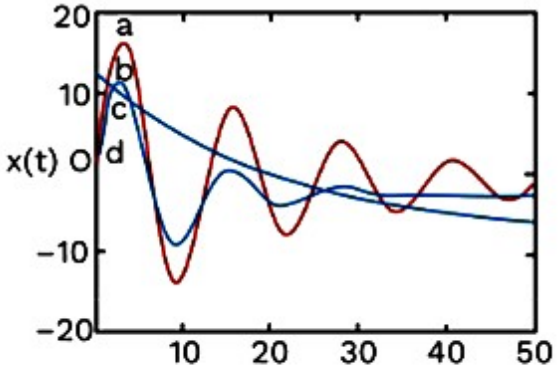
Where A is the amplitude and ω_d is the angular frequency of the damped oscillator, which is given by

$$\omega_d = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$$

The Cosine function has a period $\frac{2\pi}{\omega_d}$ but the function x(t) is not strictly periodic because of the factor $e^{-\left(\frac{bt}{2m}\right)}$, which decreases continuously with time.

The motion represented by the above equation is approximately periodic if the decrease is small in one time period T.

The solution can be graphically represented as shown in figure below. It can be regarded as a cosine function whose amplitude $\left(Ae^{-\left(\frac{bt}{2m}\right)}\right)$ gradually decreases with time.



Mechanical energy of the undamped oscillator is $\frac{1}{2}kA^2$.

Amplitude for a damped oscillating system is a function of time.

Energy of the oscillator at time t, $E(t) = \frac{1}{2}kA^2 e^{-\left(\frac{bt}{m}\right)}$

Thus, the total energy of the system decreases exponentially with time.

NOTE:

If in a damped system, the value of b is very less than $\sqrt{mk}$, then the system is said to be damped very lightly. It is generally termed as small damping.

Q. **For a damped oscillator, the mass m of the block is 400 g, k = 45 Nm^{-1} and the damping constant b is 80 g s^{-1}. Find,**
(a) The period of oscillation,
(b) Time taken for its amplitude of vibrations to drop to half of its initial value, and
(c) The time taken for its mechanical energy to drop to half of its initial value.

Sol. (a) $mk = 400 \times 10^{-3} \times 45 = 18\ kg^2\ s^{-2}$

$\therefore\ \sqrt{mk} = \sqrt{18} = 4.24\ kgs^{-1}$ Damping constant, b = 0.08 kg s^{-1} is much less than $\sqrt{mk}$. Hence, the time period T can be considered same as that of an undamped system.

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$= 2\pi\sqrt{\frac{0.4}{45}} = 0.592\ s$$

(b) Amplitude of a damped system is given by, $A_t = Ae^{-\left(\frac{bt}{2m}\right)}$

For $A_t = \frac{A}{2} = A\,e^{-\left(\frac{bt}{2m}\right)}$

$$2^{-1} = e^{-\left(\frac{bt}{2m}\right)}$$

$$\ln 2 = \frac{bt}{2m}$$

$$t = \frac{2m\ln(2)}{b} = \frac{2\times 0.4\times 0.693}{0.08} = 6.93\,s$$

(c) Mechanical energy of the damped system varies as,

$$E = E_0 e^{-\left(\frac{bt}{m}\right)}$$

For $E = \frac{E_0}{2} = E_0 e^{-\left(\frac{bt}{m}\right)}$

$$\ln 2 = \frac{bt}{m}$$

$t = \frac{m\ln(2)}{b} = \frac{0.4\times 0.693}{0.08} = 3.467\,s$ which is half time as that for amplitude.

FORCED OSCILLATIONS AND RESONANCE

When a system (like a simple pendulum is displaced from its equilibrium position and released, it oscillates with its natural frequency ω and the oscillations are called **free oscillations**.

Damping forces ultimately reduce the amplitude of the free oscillations. But, the oscillations can be maintained by applying external force. These are called forced or driven oscillations.

For the external force to be a periodic force having a frequency ωd called the driven frequency, the system oscillates at the frequency ωd of the external force (not with its natural frequency ω).

Let an external force F(t) of amplitude F_0 which varies periodically with time be applied to a damped oscillator. It can be represented as

$$F(t) = F_0 \cos\omega_d t$$

Then, by Newton's 2nd law, we can write,

$$ma(t) = -kx(t) - bv(t) + F_0 \cos\omega_d t$$

Putting $\frac{d^2x}{dt^2}$ for acceleration and rearranging it, we get

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = F_0 \cos\omega_d t$$

The natural frequency oscillations die out, when an external periodic force is applied. Now the body oscillates with the frequency of the external periodic force. Its displacement is given by

$$x(t) = A\cos(\omega_d t + \phi)$$

where t is the time measured from the moment when the periodic force is applied.

The amplitude A is a function of the forced frequency ω_d and the natural frequency ω. It is given by

$$\boxed{A = \frac{F_0}{\sqrt{\left\{m^2\left(\omega^2 - \omega_d^2\right)^2 + \omega_d^2 b^2\right\}}}}$$

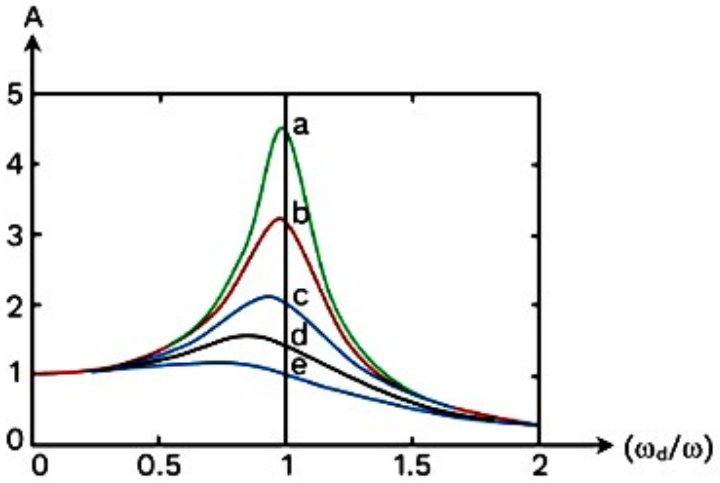
and $\boxed{\tan\phi = \frac{-v_0}{\omega_d x_0}}$

where m is the mass of the particle and v_0 and x_0 are the velocity and the displacement of the particle at time t = 0, which is the moment when the periodic force is applied. The amplitude of the forced oscillator depends on the (angular) frequency of the driving force.

Case-I Small damping, driving frequency far from natural frequency:

ω_d b is much smaller than $m(\omega^2 - \omega_d^2)$, and it can be neglected. Then amplitude becomes

$$A = \frac{F_0}{m\left(\omega^2 - \omega_d^2\right)}$$



The graph illustrate the resonant amplitude decreases with increasing damping.

Figure shows the dependence of the displacement amplitude of an oscillator on the angular frequency of the driving force for different amounts of damping.

The amplitude tends to infinity when the driving frequency equals the natural frequency. But this is the ideal case of zero damping (which never happens as damping is never perfectly zero) e.g., In a swing, when the timing of your push matches exactly with the time period of the swing, we get maximum amplitude. But, this is not infinity, as there is always some damping. This becomes clear in Case-II.

Case-II Driving frequency close to natural frequency :

If ω_d is very close to ω, $m(\omega^2 - \omega_d^2)$ is much less than $\omega_d b$, then

$$A = \frac{F_0}{\omega_d b}$$

The maximum possible amplitude for a given driving frequency is controlled by the driving frequency and the damping, and is never infinity. The phenomenon of increase in amplitude when the driving force is close to the natural frequency of the oscillator is called **resonance**.

Q. **The amplitude of vibration of a particle is given by $a_m = (a_0) / (a\omega^2 - b\omega + c)$; where a_0, a, b and c are positive. The condition for a single resonant frequency is**

(a) $b^2 = 4ac$ **(b) $b^2 > 4ac$** **(c) $b^2 = 5ac$** **(d) $b^2 = 7ac$**

Sol. (a)

For resonance, amplitude must be maximum which is possible only when the denominator of the expression is zero.

For a single resonant frequency, $b^2 = 4ac$